1. Cover Page

COMP 3613

Assignment 2

MyAdvisor

Project, Test Plans, Implementation and Reports

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Submission date	27 October 2026
Repository	https://github.com/Camcoon/MyAdvisorWithTests

Submission dossier: core lecturer deliverables are assembled. Historical Postman/UAT results are explicitly identified; publication of the latest local changes and corrected-build manual/API verification remain pending (Appendix B).

Identity verified against the retained student statement and Assignment 1 cover. The eight-digit Student ID in the original wireframe and browser evidence is demonstration data.

This report distinguishes planned outcomes, historical executions and corrected-build regression evidence. It does not certify unperformed tests.

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2. Project Description

MyAdvisor is a coursework prototype for Student–Advisor academic planning. It brings degree progress, prerequisite-aware course selection and plan decisions into connected workspaces, addressing the difficulty of reconciling completed work, current courses and proposed semester choices across separate records.

Workflow / user	Purpose and implemented behavior
WF1: Track Degree Progress — Student	View completed, in-progress and remaining credits; overall progress; requirement categories; recent academic activity.
WF2: Plan Semester Course Selections — Student	Search/filter the catalogue, inspect eligibility, review and submit selections, read feedback, revise returned plans, remove permitted courses and cancel/replace plans.
WF3: Review and Approve Course Plans — Advisor	Inspect the shared Pending queue and Student/course details; Approve, Reject or Request Changes, with optional feedback.

Students use Home, Degree Progress, Plan Courses, Messages and Profile. Advisors use their own Home/Profile and Advisor Review. Student identity comes from the authenticated account; Advisor decisions update a plan’s status, reviewer and feedback. Semesters use the nine fixed Year 1–3 / semester 1–3 options, with semester 3 designated summer.

Technology	Role
Python / FastAPI / Pydantic	HTTP application, request validation and JSON APIs.
SQLModel / SQLAlchemy / SQLite	Models, repository persistence and isolated local tests. The Render blueprint targets PostgreSQL; no production database test is claimed.
Jinja2 / Bootstrap / AdminLTE / JavaScript	Rendered Student/Advisor pages and selection/review interactions.
pwdlib / PyJWT / signed cookies	Password verification, signed login token, current-role and ownership checks.
pytest / TestClient / mocks / Postman	Isolated unit decisions, integrated HTTP/database checks and automated API collection.

Assignment 2 scope. The Assignment 1 workflows are extended with explicit JSON API contracts, unit/integration plans and pytest tests, a 90-request Postman collection, manual UAT and traceability. The owner-approved fictional demonstration curriculum totals 93 non-overlapping credits: Core 36, General Education 18, Electives 12, Major 18 and Additional 9. It is not an official UWI curriculum.

Source: docs/report.md; app/models; app/routers; docs/assignment2/api.md.

3. Use Case Diagram

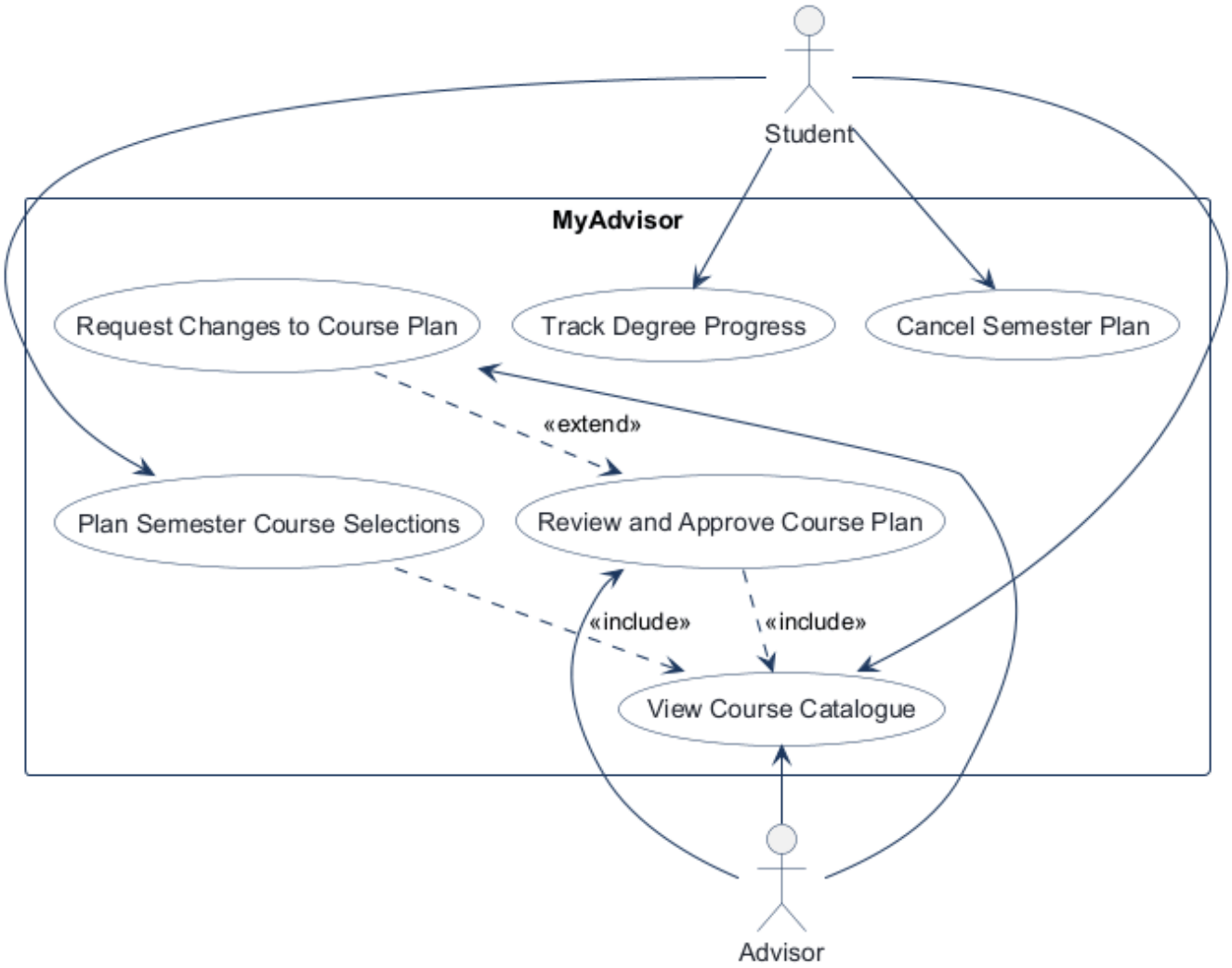


Figure 3.1. Original Assignment 1 MyAdvisor use case diagram, reproduced from docs/diagrams/use-case.png without redrawing.

The Student tracks degree progress, plans semester selections, views the catalogue and cancels semester plans. The Advisor reviews course plans and can request changes. Catalogue viewing is included in the planning/review workflows; Request Changes extends review. These are the original diagram’s actors and relationships. Authentication supports their implementation and is tested separately.

4. Entity Relationship Diagram

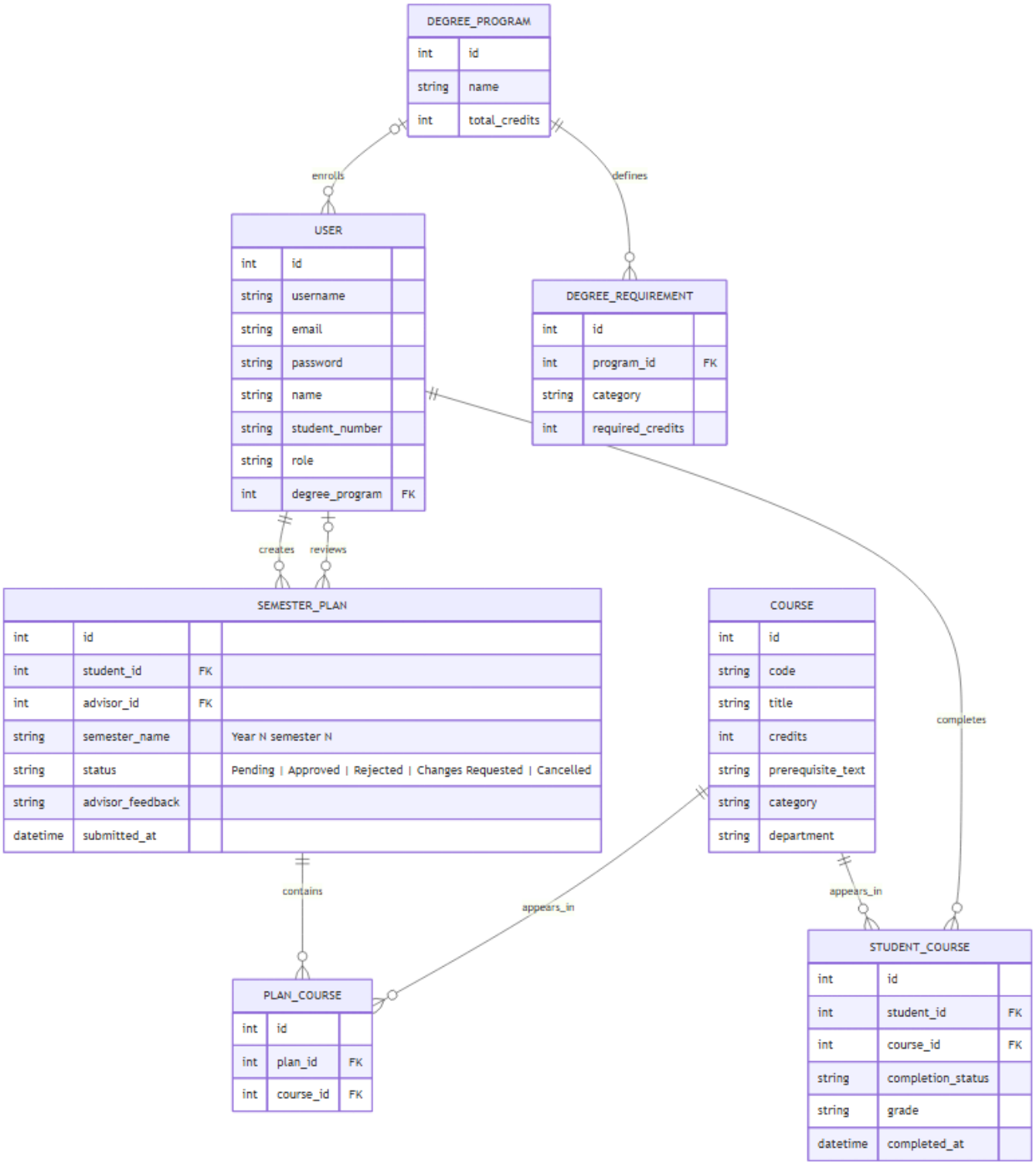


Figure 4.1. Original MyAdvisor ERD image extracted losslessly from Assignment 1 docs/report.pdf, page 3 (image 29). The readable keyed rendering follows.

A program defines requirements and optionally enrolls users. A Student creates plans and has course records; an optional Advisor reviews each plan. PLAN_COURSE joins plans to courses; STUDENT_COURSE joins users to academic course records. All seven id fields are primary keys. Unique pairs constrain program/category, student/course and plan/course. The password field names stored data; no value is printed.

4.1 Readable entity keys and relationships

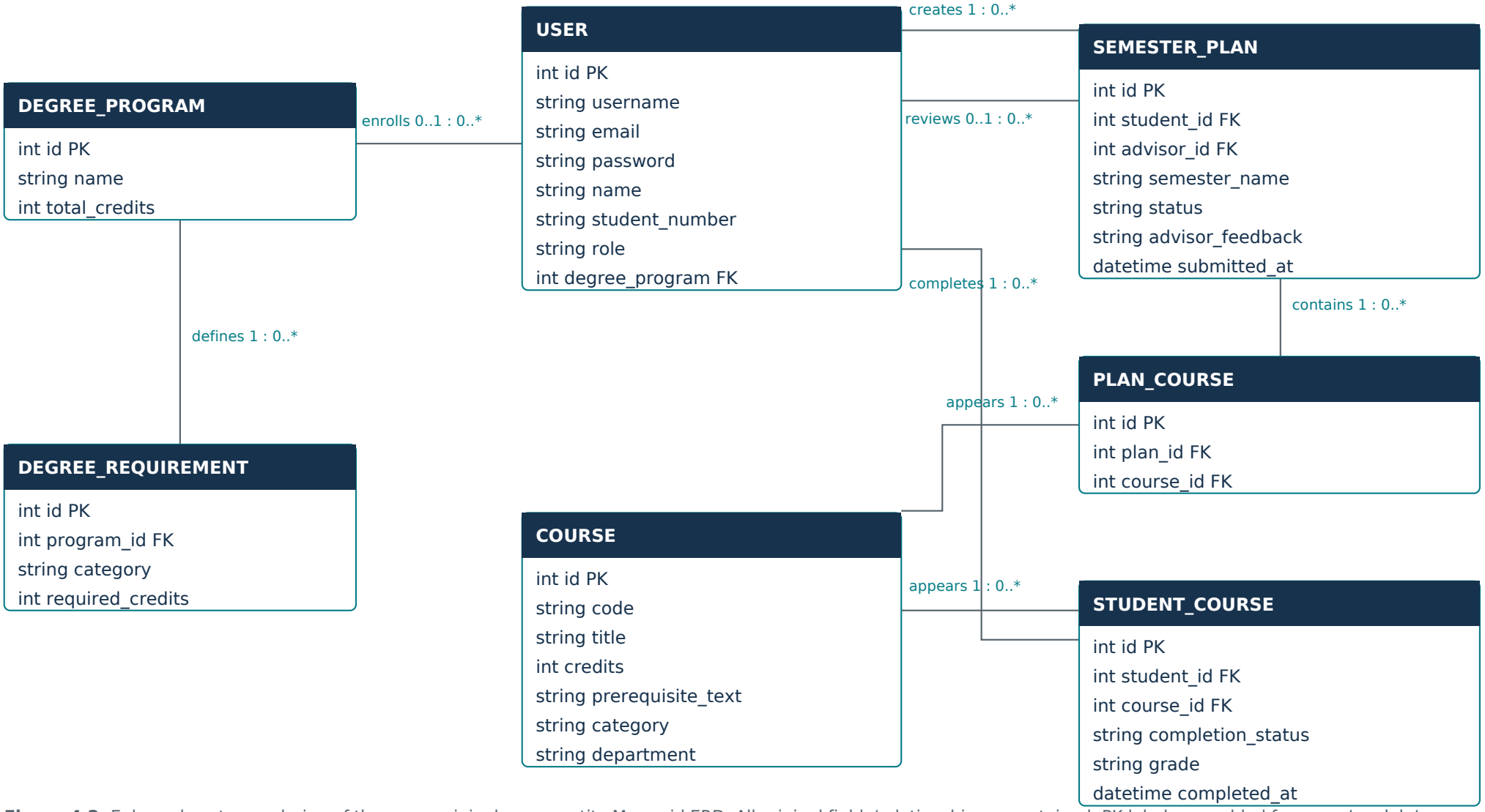


Figure 4.2. Enlarged vector rendering of the same original seven-entity Mermaid ERD. All original fields/relationships are retained; PK labels are added from app/models/user.py and app/models/advising.py. FK fields retain their original names. Numeric edge labels state source:target cardinality; “appears” abbreviates “appears_in”.

5. Application Wireframe

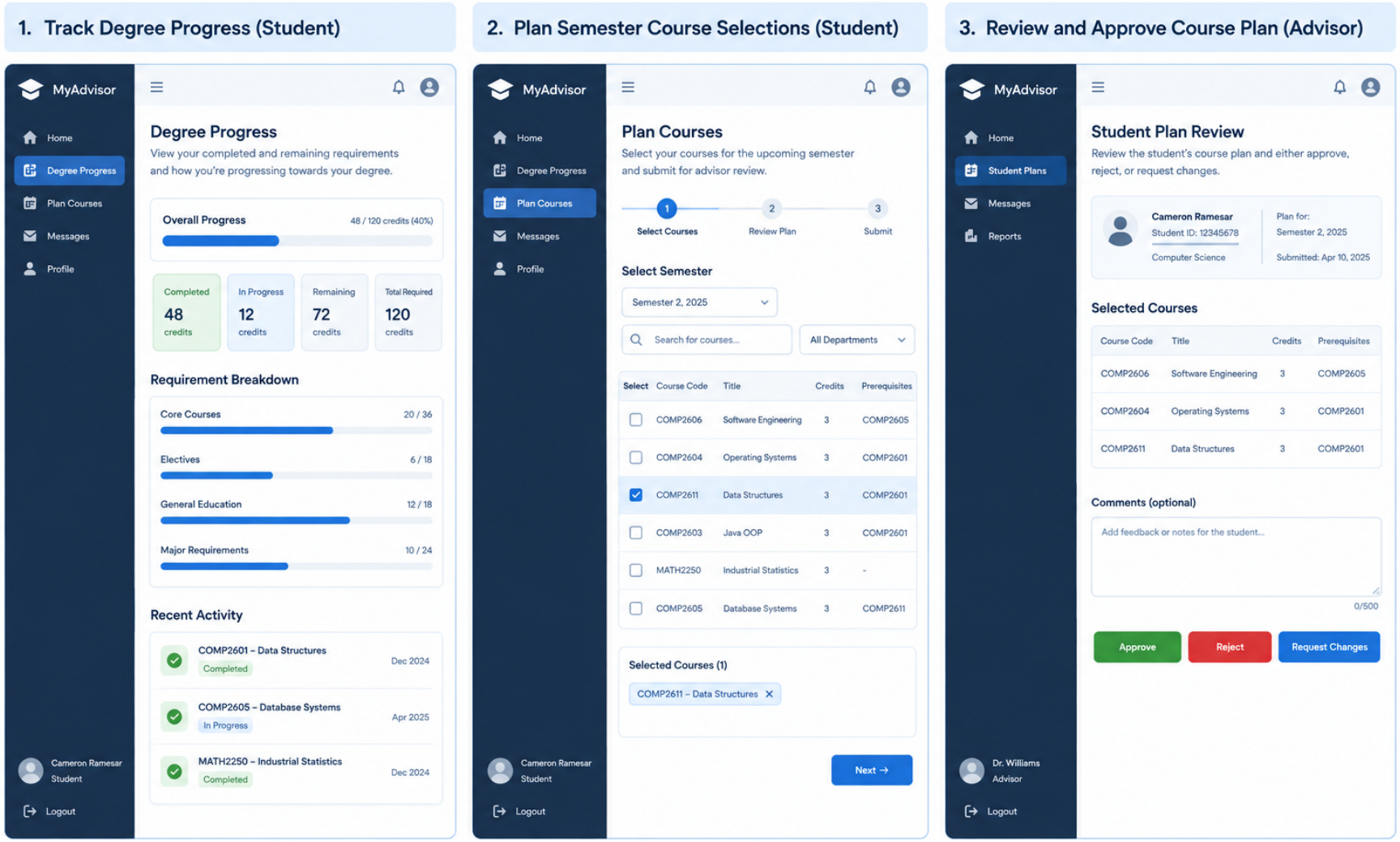


Figure 5.1. Original three-panel Assignment 1 wireframe, unmodified. W1: Student Degree Progress; W2: Student Semester Planning; W3: Advisor Plan Review. Sample credits, dates, course labels and the eight-digit ID are illustrative, not execution results.

5.1 Wireframe detail: W1 — Student Degree Progress

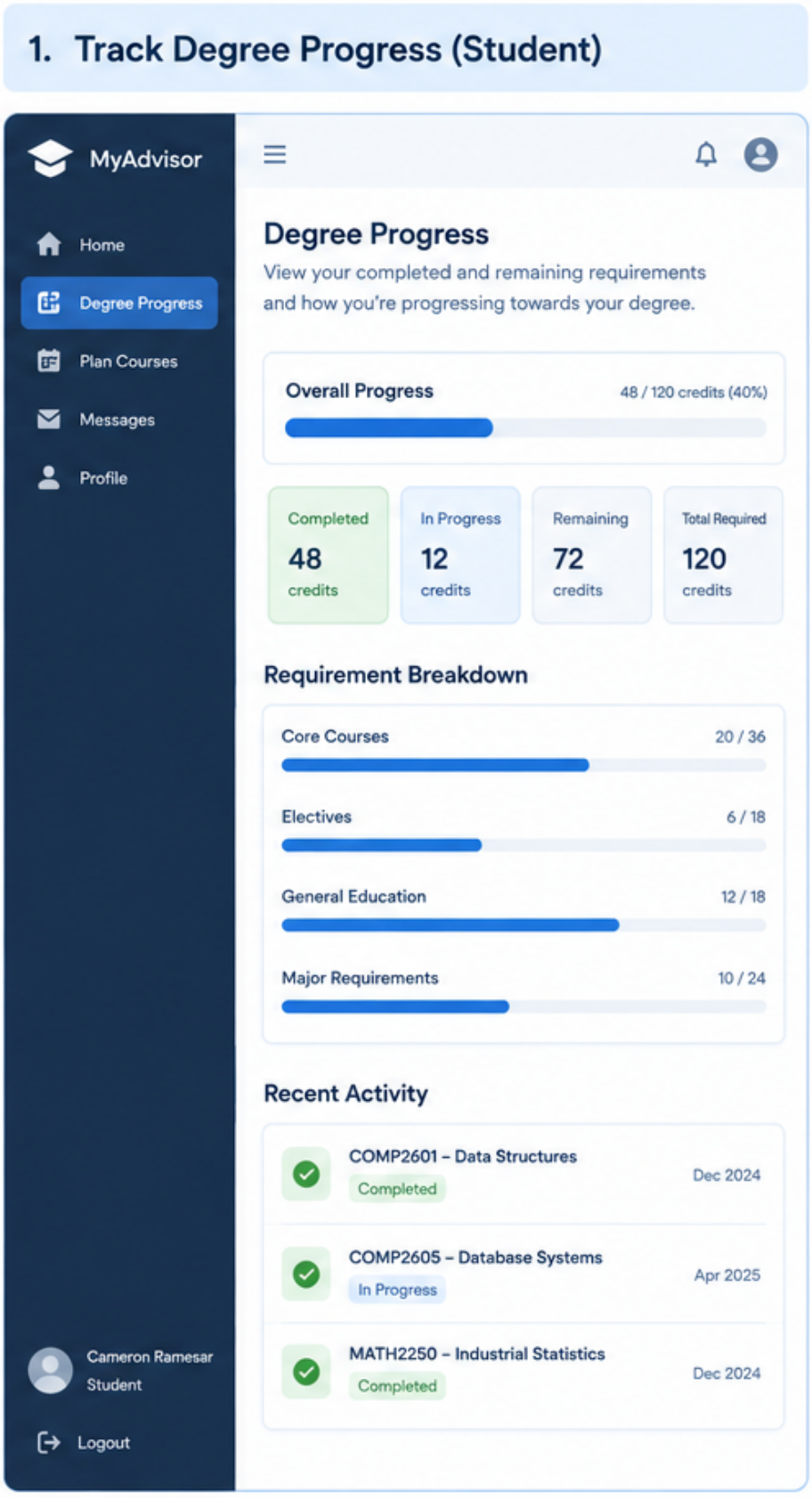


Figure 5.2. Enlarged crop of the original panel. W1 maps to UAT-01; W2 to UAT-02–05 and UAT-08–17; W3 to UAT-06–11 and queue checks. Login and cancellation extensions are documented in the plan.

5.2 Wireframe detail: W2 — Student Semester Planning

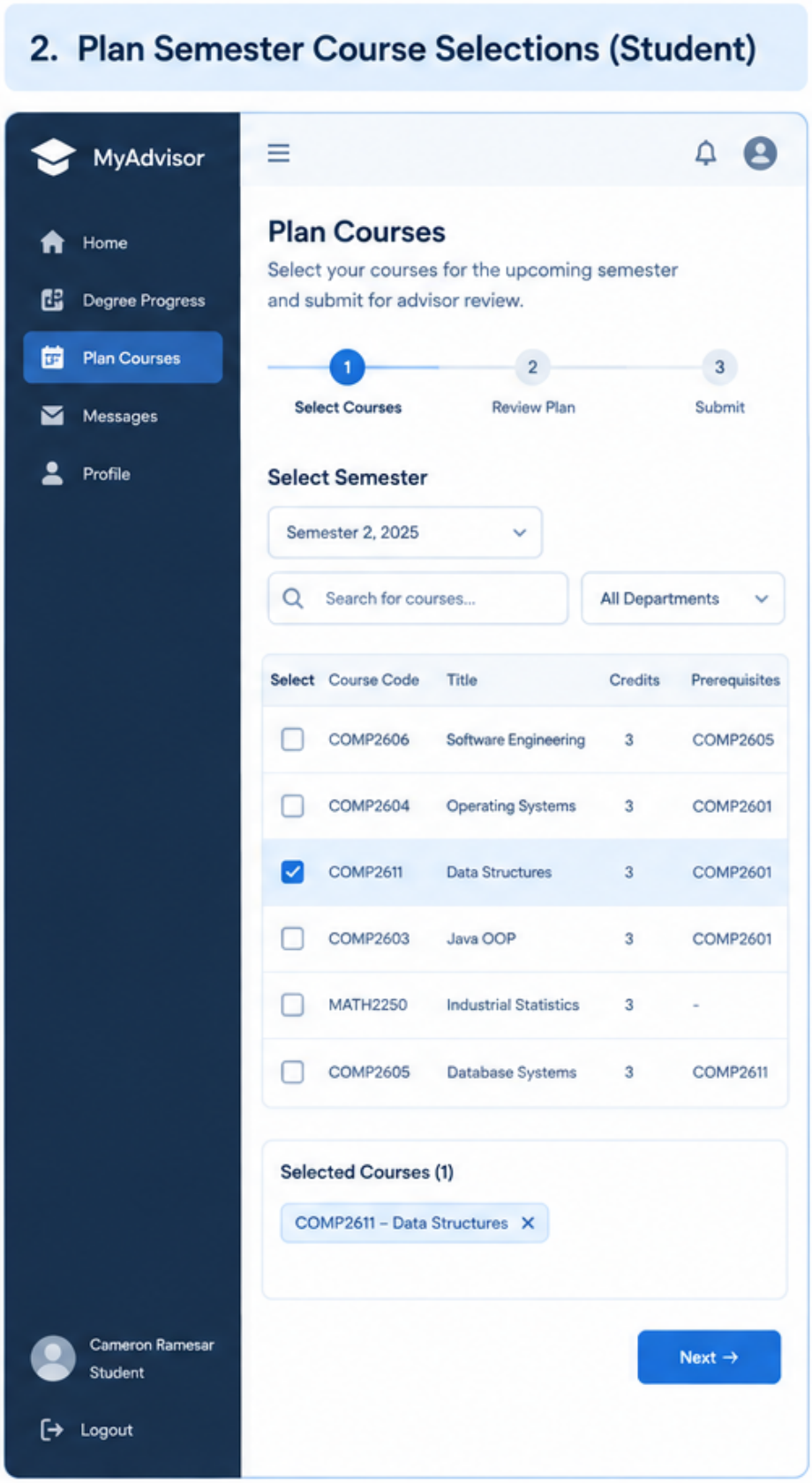


Figure 5.3. Enlarged crop of the original panel. W1 maps to UAT-01; W2 to UAT-02–05 and UAT-08–17; W3 to UAT-06–11 and queue checks. Login and cancellation extensions are documented in the plan.

5.3 Wireframe detail: W3 — Advisor Plan Review

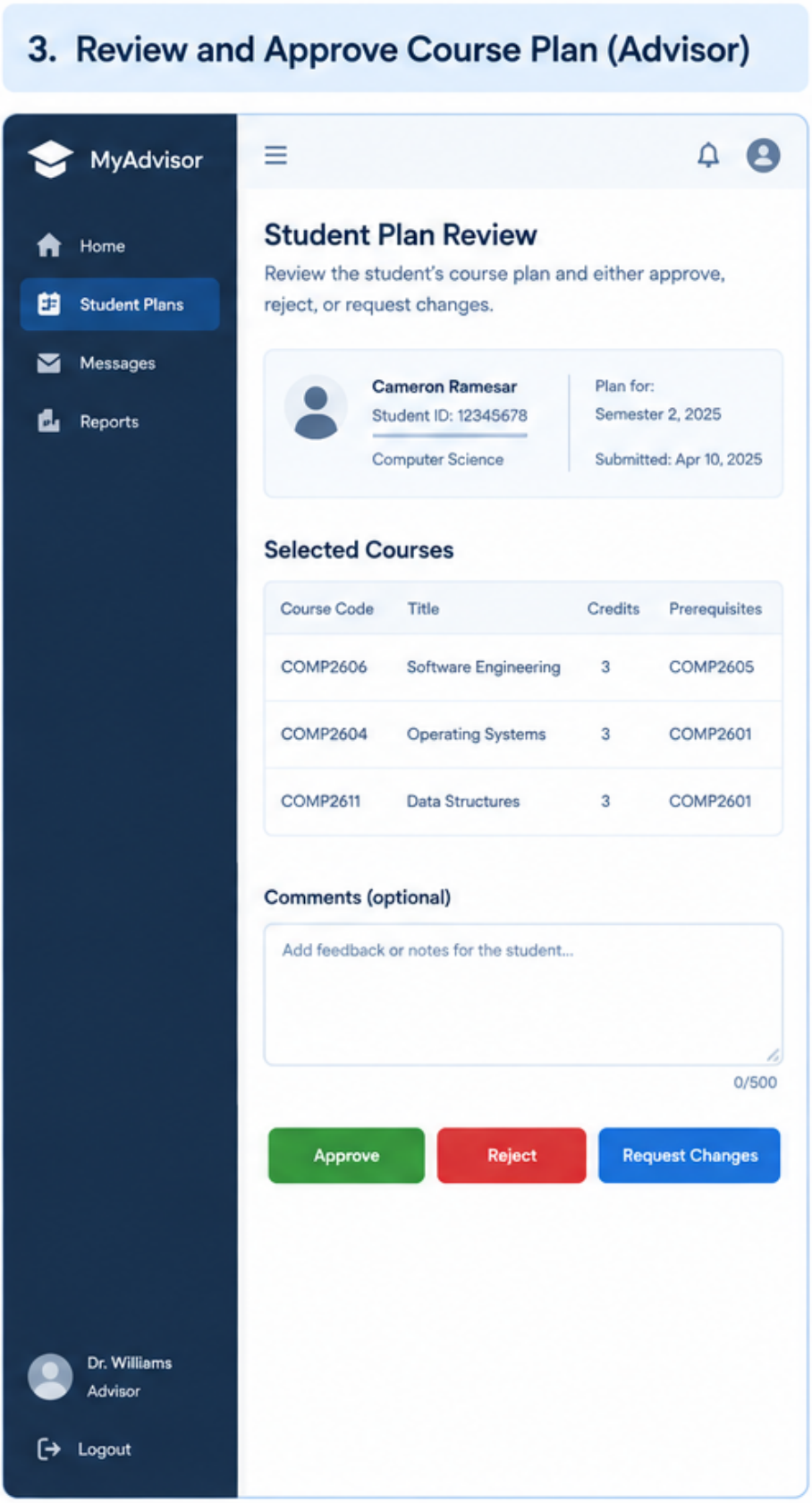


Figure 5.4. Enlarged crop of the original panel. W1 maps to UAT-01; W2 to UAT-02–05 and UAT-08–17; W3 to UAT-06–11 and queue checks. Login and cancellation extensions are documented in the plan.

6. Test Plans

6.1 User Acceptance Test Plan

Objective and scope. Verify that Students can understand degree progress and complete eligible semester planning, and Advisors can review and return decisions. These are planned browser tasks and expected outcomes. Actual results and latest statuses appear only in Appendix A; automated executions appear in Section 8.

Template access. Both lecturer Google Docs templates were requested through their edit and text-export URLs on 27 October 2026; the tool reported them inaccessible. Their exact headings, extra fields, sign-offs and formatting are **unverified**, and are not invented. This section adapts the retained plans to the fields explicitly required in the supplied brief. Test Plans template; User Acceptance Tests Plan template.

Plan control	Specification
Actors	Student (regular_user), Advisor (admin); separate browser profiles to isolate cookies.
Environment	Prepared disposable SQLite database; current local app; JavaScript enabled; planned base URL http://127.0.0.1:5002. Record actual URL, build/source hashes, browser/version, OS, date/time zone and tester at execution. Existing historic browser/build details are incomplete.
Preparation	Use the existing safe setup in reports/uat-execution.md and the corrected-build retest guide. Do not reset an existing database. Confirm empty A/B/C semesters and expected demo identities. Use supplied account credentials privately; no password values appear here.
Execution order	UAT-18 → UAT-01–05 → UAT-19 → UAT-06–17 → UAT-20. Numeric IDs are retained.
References	W1/W2/W3 are the panels in Figures 5.2/5.3/5.4. W4 means Student Messages/feedback; W5 means cancellation/resubmission extensions to W2/W3, not additional original panels.
Recording and completion	Record observed outcome, evidence, deviation and defect for every step. Pass only when the complete expected outcome is observed. Partial evidence remains Not Run; missing evidence is not an invented failure.

6.1.1 Planned UAT data and baseline

Symbol / item	Planned data
A / B / C	Year 1 semester 1 / Year 1 semester 2 / Year 1 semester 3.
E1 / E2 / E3	COMP 1201 Object-Oriented Programming (requires completed COMP 1101) / MATH 2250 Industrial Statistics / ENGL 1000 Academic Writing. Each is 3 credits; E2/E3 have no prerequisites.
Academic records	COMP 1101 and MATH 1200 Completed, 3 credits each; COMP 1600 In Progress, 3 credits. COMP 2100 is blocked because COMP 1600 is not completed.
Degree / categories	93 total; 6 completed; 3 in progress; 84 remaining; 6% completed. Approved fictional category requirements: Core 3/36 (33 remaining), General Education 3/18 (15), Electives 0/12 (12), Major 0/18 (18), Additional 0/9 (9). Category targets sum to 93 with no overlap. Historical screen captures show older values.

The following case specifications come from the latest execution companion’s planned fields. They preserve the original two-course inputs and expected results; the historical three-course deviation is recorded separately. Source-only stale Not Run/Passed fields are excluded from the plan.

UAT-01 — View degree progress

Field — UAT-01	Planned specification
Actor / wireframe	Student; W1: Overall Progress, credit cards, Requirement Breakdown, Recent Activity.
Objective	Understand completed, current, and remaining degree requirements.
Preconditions	Local setup and UAT-18 completed; Student signed in; compare displayed data with the documented seed baseline.
Test data	Student demo account; baseline progress and category values above.
Steps	1. In the signed-in Student window click **Degree Progress** in the sidebar. 2. Read Overall progress, all four credit cards, every Requirement breakdown row, and Recent course activity. Capture the displayed values. 3. Refresh the page and compare the values again.
Expected result	Student workspace opens. Credits are 6 completed, 3 in progress, 84 remaining, and 93 required; overall progress is 6 / 93 (6%). Breakdown matches the baseline and distinguishes completed from in-progress courses. Recent activity shows the three recorded courses with their correct statuses. Refresh preserves the values. Record the category-total inconsistency noted above for acceptance review.

UAT-02 — Browse, search, and filter available courses

Field — UAT-02	Planned specification
Actor / wireframe	Student; W2: semester dropdown, search, department filter, course list.
Objective	Find courses and understand their credits, prerequisites, and eligibility.
Preconditions	UAT-01 completed; Student signed in; A has no plan.
Test data	A; E1/E2; search COMP 1201, Industrial Statistics, and ZZZ-UAT-NO-COURSE.
Steps	1. Click **Plan Courses**. Open **Semester** and inspect the nine Year 1–3/semester 1–3 options; choose **Year 1 semester 1**. 2. Scroll the course list and inspect course codes, titles, credits, departments, prerequisites, and eligibility messages. 3. Enter COMP 1201 in **Search by code or course name**; inspect the matching row. 4. Replace the search with Industrial Statistics; inspect MATH 2250. 5. Clear the search and choose **Mathematics** in the department dropdown; inspect the remaining rows. 6. With Mathematics still selected, search COMP 1201; inspect the no-match message, then click **Clear Filters**. 7. Search ZZZ-UAT-NO-COURSE; inspect the no-match message and click **Clear Filters** again.
Expected result	The chosen semester loads, and semester 3 is identified as summer. E1 and E2 show their documented details. Search matches code or title, and the department filter restricts results. Combined incompatible filters and the nonexistent search show “No courses match your search and department filters.” Clear Filters restores the catalog and All departments. No plan is submitted.

UAT-03 — Select only eligible courses and review the selection

Field — UAT-03	Planned specification
Actor / wireframe	Student; W2: checkboxes, Selected Courses, Next/Review Plan.
Objective	Prevent an empty or ineligible selection and check the proposed credit load.
Preconditions	UAT-02 completed; A selected; filters clear; no boxes selected.
Test data	E1 and E2 (6 credits); ineligible baseline examples.
Steps	1. Observe **0 courses selected** and attempt to click **Review Selected Courses**. 2. Locate COMP 1101, COMP 1600, and COMP 2100, scrolling as necessary. Read their eligibility messages and attempt to select each checkbox. 3. Select COMP 1201 and MATH 2250. Check the count, then click **Review Selected Courses**. 4. Read the two reviewed courses, their prerequisites, and **Total selected credits**. 5. Click **Back to course selection**, uncheck MATH 2250, and check the count. Recheck it and click **Review Selected Courses** again.
Expected result	Empty selection cannot advance. The three ineligible checkboxes are disabled with the documented reasons; an in-progress prerequisite does not count as completed. Only E1/E2 are selected. The count moves 0 → 2 → 1 → 2; review shows exactly E1/E2 and 6 credits. Returning to selection retains choices. Nothing is submitted yet; leave the browser on review for UAT-04.

UAT-04 — Submit semester plan A

Field — UAT-04	Planned specification
Actor / wireframe	Student; W2: Review Plan and Submit.
Objective	Send the reviewed selection to an Advisor.
Preconditions	UAT-03 left E1/E2 on the review step for A; no existing A plan.
Test data	A; E1/E2, 6 credits.
Steps	1. Confirm E1/E2 and 6 credits on **Review your selection**. 2. Click **Submit Plan for Advisor Review** once. 3. Read the confirmation, semester/status banner, and **Selected courses for Year 1 semester 1** list. 4. Refresh the page. Attempt to alter a course checkbox and click **Review Selected Courses**.
Expected result	“Semester plan submitted for advisor review.” appears. A is Pending with exactly E1/E2 and is awaiting advisor review. Refresh preserves the submission. Standard selection controls and the review button are disabled for the submitted semester, preventing another submission through this form. Pending-plan Remove/Cancel controls remain available as separate permitted actions; leave them untouched here.

UAT-05 — View submission status

Field — UAT-05	Planned specification
Actor / wireframe	Student; W2/W4: submitted plan, Home, Messages navigation.
Objective	Find the current submission state after navigating away and signing in again.
Preconditions	UAT-04 completed; only A has been submitted.
Test data	A Pending; E1/E2, 6 credits.
Steps	1. Click **Home** and read **Latest semester plan**. 2. Click **Messages** and find the **Year 1 semester 1 plan** entry. 3. Click **Sign out**, open /login, enter the Student credentials, and click **Sign in**. 4. Click **Plan Courses**, choose **Year 1 semester 1**, and inspect its status and selected courses.
Expected result	Home shows A Pending, 2 courses, 6 credits, and a waiting-for-review message. Messages identifies the same semester as Pending. Signing in again preserves A and its selection; no decision is implied before the Advisor acts.

UAT-06 — Advisor reviews the submitted plan

Field — UAT-06	Planned specification
Actor / wireframe	Advisor; W3: Student Plans, student summary, Selected Courses table.
Objective	Inspect the correct Student's submission and credit load before deciding.
Preconditions	UAT-05 and UAT-19 completed; A Pending; Advisor signed in in its separate browser profile.
Test data	Advisor demo account; Student identity recorded during setup; A; E1/E2.
Steps	1. Confirm the separate Advisor window shows Dr. Williams and Advisor workspace after UAT-19. 2. Click **Advisor Review** in the sidebar. 3. Find the card matching the recorded Student identity and **Year 1 semester 1**. 4. Inspect student ID, program, submitted date, Pending status, course count, each course row, prerequisites, and **Total requested credits**. 5. Locate **Comments (optional)**, its 0/500 counter, **Approve Plan**, **Reject Plan**, and **Request Changes**. Do not click a decision yet.
Expected result	The Advisor sees A in the queue with the correct Student identity, Computer Science program, submission date matching execution, E1/E2, 2 courses, and 6 total credits. All three decision controls are present. Reading the card leaves the plan Pending. Match by Student plus semester because the queue may contain other students.

UAT-07 — Request changes and deliver feedback

Field — UAT-07	Planned specification
Actor / wireframe	Advisor (Student verifies delivery in the named Student steps); W3: Comments and Request Changes; W4/W5: feedback and returned plan.
Objective	Return a plan with clear revision instructions that the Student can read.
Preconditions	UAT-06 completed; A still Pending.
Test data	Feedback: Replace MATH 2250 with ENGL 1000, then resubmit.
Steps	1. In the Advisor's A card, enter the exact feedback above into **Comments (optional)**. Observe the character counter. 2. Click **Request Changes** and read the confirmation; refresh **Advisor Review**. 3. Click Advisor **Messages** and locate the Student/A decision. 4. In the Student window click **Messages**, refreshing if already there, and locate A. 5. Read the feedback and click **Review and update plan**. Inspect the semester, status, selected courses, and available actions.
Expected result	Advisor confirmation says changes were requested. A disappears from the Pending queue and appears in Advisor Messages as Changes Requested with the entered feedback. The Student sees the same status/feedback and opens editable A with E1/E2 preselected. **Cancel Plan** is absent for Changes Requested; the instruction to update and resubmit is visible. No approval or rejection has occurred.

UAT-08 — Modify and resubmit the returned plan

Field — UAT-08	Planned specification
Actor / wireframe	Student (Advisor verifies the queue in the named Advisor steps); W2/W5: editable selection, Review Plan, resubmission; W3: revised queue card.
Objective	Apply the requested change and return the revised selection for review.
Preconditions	UAT-07 completed; Student viewing editable A with E1/E2 selected.
Test data	Keep E1; replace E2 with E3; revised total 6 credits.
Steps	1. In Student **Plan Courses** for A, uncheck MATH 2250 and check ENGL 1000, scrolling the list as necessary. 2. Click **Review Selected Courses** and confirm COMP 1201 plus ENGL 1000 and 6 credits. 3. Click **Resubmit Plan for Advisor Review** and inspect the resulting status and selected-course list. 4. Click Student **Messages** and inspect A's current update. 5. In the Advisor window click **Advisor Review**, refresh, and inspect the Student/A card.
Expected result	A returns to Pending with exactly E1/E3. The old request-changes feedback is cleared on resubmission; Messages reflects the current Pending state rather than promising immutable feedback history. Advisor queue contains one A card for this Student with the revised courses and 6 credits, ready for another decision. Capture the previous feedback in UAT-07 before this step replaces it.

UAT-09 — Approve the revised plan

Field — UAT-09	Planned specification
Actor / wireframe	Advisor (Student verifies delivery in the named Student steps); W3: Comments and Approve; W2/W4: plan status and Messages.
Objective	Approve the revised plan and let the Student see the decision.
Preconditions	UAT-08 completed; revised A Pending.
Test data	A; E1/E3; feedback Revised selection approved.
Steps	1. In Advisor **Advisor Review**, find the Student/A card and confirm E1/E3. 2. Enter Revised selection approved. into its **Comments (optional)** and click **Approve Plan**. 3. Read the confirmation and refresh the review queue. Click Advisor **Messages** and inspect A. 4. In the Student window click **Messages** and read A's status/feedback. 5. Click **Plan Courses**, choose A, and inspect the selected courses and editing controls. Refresh once.
Expected result	“Plan approved.” appears and A leaves the Pending queue. Both actors see Approved and the exact feedback. Student A retains E1/E3 and remains Approved after refresh. Course checkboxes/review button are locked; Pending-only Remove buttons are absent. **Cancel Plan** remains available because the current implementation permits cancellation of Approved plans; do not cancel until UAT-14.

UAT-10 — Submit a separate plan for rejection

Field — UAT-10	Planned specification
Actor / wireframe	Student; W2: semester, selection, review, submit.
Objective	Submit a different semester without altering the approved plan.
Preconditions	UAT-09 completed; A Approved; B has no plan.
Test data	B = Year 1 semester 2; E2 only, 3 credits.
Steps	1. Click Student **Plan Courses** and choose **Year 1 semester 2**. 2. Select only MATH 2250; click **Review Selected Courses** and confirm one course, 3 credits. 3. Click **Submit Plan for Advisor Review** and inspect B's status/list. 4. Choose **Year 1 semester 1** to inspect A, then switch back to **Year 1 semester 2**.
Expected result	B is Pending with only E2; A remains Approved with E1/E3. Switching semesters displays the selected semester's own plan and status without mixing the two selections.

UAT-11 — Reject the separate plan

Field — UAT-11	Planned specification
Actor / wireframe	Advisor (Student verifies delivery in the named Student steps); W3: Comments and Reject; W4/W5: rejected status and revision option.
Objective	Reject B and communicate the reason to the Student.
Preconditions	UAT-10 completed; B Pending.
Test data	B; feedback Plan rejected for this UAT scenario; choose a different course mix.
Steps	1. In the Advisor window click **Advisor Review**, refresh, and find the Student's **Year 1 semester 2** card containing only MATH 2250. 2. Enter the exact feedback above into that card's **Comments (optional)** and click **Reject Plan**. 3. Read the confirmation, refresh the queue, then inspect B in Advisor **Messages**. 4. In the Student window click **Messages** and inspect B's status and feedback. 5. Click B's **Review and update plan** link; inspect the editable course selection and **Cancel Plan** control without using either yet.
Expected result	“Plan rejected.” appears; B leaves the Pending queue. Both actors see Rejected with the exact reason. Student B is editable for resubmission and cancellation is available. A's approval remains unchanged. Leave B Rejected for UAT-15.

UAT-12 — Submit a pending plan for cancellation checks

Field — UAT-12	Planned specification
Actor / wireframe	Student (Advisor verifies the queue in the named Advisor steps); W2: semester, selection, review, submit; W3: queue.
Objective	Prepare a Pending plan through normal browser submission.
Preconditions	UAT-11 completed; C has no plan.
Test data	C = Year 1 semester 3; E1/E2, 6 credits.
Steps	1. Click Student **Plan Courses**, choose **Year 1 semester 3**, and select COMP 1201 and MATH 2250. 2. Click **Review Selected Courses**, verify 6 credits, and click **Submit Plan for Advisor Review**. 3. Inspect C's Pending status and its selected-course list. 4. In the Advisor window click **Advisor Review**, refresh, and find the Student/C card; make no decision.
Expected result	C is Pending with E1/E2 and is visible to the Advisor with 6 credits. A remains Approved and B remains Rejected. C stays Pending for UAT-13 and UAT-16.

UAT-13 — Remove a course from a Pending plan and protect the last course

Field — UAT-13	Planned specification
Actor / wireframe	Student (Advisor verifies the queue in the named Advisor steps); W2/W5: selected-course removal; submitted-plan Remove is an implementation extension.
Objective	Adjust a submitted Pending plan without leaving an empty plan.
Preconditions	UAT-12 completed; C Pending with E1/E2.
Test data	Remove E2; retain E1, 3 credits.
Steps	1. In Student **Plan Courses**, choose C and click **Remove** beside MATH 2250 in **Selected courses for Year 1 semester 3**. 2. Dismiss the browser confirmation with **Cancel** and verify both courses remain. 3. Click the same **Remove** again and accept the confirmation with **OK**. 4. Read the confirmation and inspect the remaining course and available removal action. Refresh. 5. Refresh Advisor **Advisor Review** and inspect C's course table and total.
Expected result	Dismissing the dialog changes nothing. Confirming removes only E2 and shows “Course removed from the submitted plan.” C stays Pending with E1 and 3 credits. The remaining course has no Remove button and displays “Cancel the plan to remove its last course.” Advisor queue reflects E1/3 credits after refresh.

UAT-14 — Cancel an Approved plan

Field — UAT-14	Planned specification
Actor / wireframe	Student; W5: cancellation extension to panel 2.
Objective	Cancel A using the implemented permission for Approved plans.
Preconditions	UAT-09 approval retained; A Approved; UAT-12/13 completed.
Test data	A; Approved → Cancelled.
Steps	1. Click Student **Plan Courses**, choose A, and confirm its current status is Approved. 2. Click **Cancel Plan** and dismiss the confirmation with **Cancel**; verify A remains Approved with E1/E3. 3. Click **Cancel Plan** again and accept the confirmation with **OK**. 4. Read the success message; refresh and inspect status, selected courses, and available actions. 5. Click **Messages** and find A's status.
Expected result	Dismissing cancellation leaves approval unchanged. Confirming shows “Semester plan cancelled. You can edit it and resubmit.” A becomes Cancelled and remains so after refresh; Messages also shows Cancelled. The planning screen clears the active selection, permits a new choice, and hides Cancel Plan. Previous feedback may remain attached to the cancelled record; it must not make the current status appear Approved.

UAT-15 — Cancel a Rejected plan

Field — UAT-15	Planned specification
Actor / wireframe	Student; W5: cancellation extension to returned plan.
Objective	Cancel B instead of resubmitting the rejected selection.
Preconditions	UAT-11 rejection retained; B Rejected.
Test data	B; Rejected → Cancelled.
Steps	1. Click Student **Plan Courses** and choose B; confirm Rejected and its rejection feedback. 2. Click **Cancel Plan** and accept the confirmation with **OK**. 3. Read the success message, refresh, and inspect the status, selection count, and actions. 4. Click **Messages** and find B's entry.
Expected result	B becomes Cancelled and persists after refresh. No active courses are selected; a new selection is possible and Cancel Plan is absent. Messages reports Cancelled. A's cancellation and C's Pending state are unaffected. Retained rejection feedback does not override the current Cancelled status.

UAT-16 — Cancel a Pending plan and remove it from the review queue

Field — UAT-16	Planned specification
Actor / wireframe	Student (Advisor verifies the queue in the named Advisor steps); W5: Cancel Plan; W3: Pending review queue.
Objective	Withdraw C before the Advisor makes a decision.
Preconditions	UAT-13 completed; C Pending with E1; no Advisor decision on C.
Test data	C; Pending → Cancelled.
Steps	1. Click Student **Plan Courses**, choose C, and confirm Pending with only COMP 1201. 2. Click **Cancel Plan** and accept the confirmation with **OK**. 3. Read the success message and refresh; inspect the Cancelled banner, selection count, and available actions. 4. Click Student **Messages** and inspect C's status. 5. In the Advisor window click **Advisor Review**, refresh, and look for the Student/C card.
Expected result	C becomes Cancelled, the active selection clears, and no repeat cancellation control appears. Student Messages shows Cancelled. The Student/C card is absent from the refreshed Pending queue, so it cannot be reviewed there. Other students' Pending submissions, if any, remain available.

UAT-17 — Start a new plan after cancellation

Field — UAT-17	Planned specification
Actor / wireframe	Student (Advisor verifies the queue in the named Advisor steps); W2/W5: new selection after cancellation, review and submit; W3: queue.
Objective	Submit a fresh selection for the same semester after withdrawing its previous plan.
Preconditions	UAT-16 completed; C Cancelled with no active selection.
Test data	C; E3 only, 3 credits.
Steps	1. Click Student **Plan Courses**, choose C, and confirm no course is selected. 2. Select only ENGL 1000 and click **Review Selected Courses**. 3. Confirm ENGL 1000 and 3 credits, then click **Resubmit Plan for Advisor Review** (the implemented label on the cancelled-plan screen). 4. Inspect the resulting Pending status and selected-course list; refresh once. 5. In the Advisor window click **Advisor Review**, refresh, and inspect the Student/C card.
Expected result	C's current plan is Pending with only E3 and 3 credits; the previously cancelled E1 selection does not reappear. Advisor queue contains one active Pending C card for this Student with E3. The screen describes this as a brand new plan even though the action label says Resubmit. Historical Cancelled entries may remain in Messages. Do not infer database identifiers from the browser or require deletion of history.

UAT-18 — Student login and clean-run baseline (execute first)

Field — UAT-18	Planned specification
Actor / wireframe	Student; Student identity/sidebar in panels 1–2; login form is outside the original wireframe
Objective	Verify Student login, identity, empty baseline and persistent session.
Preconditions	Safe local setup complete; fresh database; dedicated Student browser profile signed out; server running
Test data	[Student test account] / [Student test password]; invalid password [invalid Student test password]; semesters A/B/C
Steps	1. In the Student window open http://127.0.0.1:5002/login. 2. Enter [Student test account] in **Username**, [invalid Student test password] in **Password**, then click **Sign in**. Inspect the response. 3. Enter [Student test account] and [Student test password] and click **Sign in**. 4. Confirm **Student workspace**, Home, Cameron Ramesar and Student ID 12345678/Computer Science. Click **Profile** if needed to read **Username**, **Student ID** and degree program. 5. Click **Home** and confirm the no-submitted-plan message in **Latest semester plan**. Click **Messages** and confirm **No advisor updates yet**. 6. Click **Plan Courses**. Choose **Year 1 semester 1**, then **Year 1 semester 2**, then **Year 1 semester 3**, waiting for each reload. Confirm each has zero selected courses and no existing plan status banner or selected-course list. Return to **Year 1 semester 1**. 7. Click **Home** and refresh. Confirm the Student remains signed in. Continue with UAT-01.
Expected result	Invalid credentials show **Incorrect username or password** on the login page without opening the workspace. Valid credentials open /app as the Student, with the seeded identity, Student navigation and no Advisor Review link. Home, Messages and A/B/C have the fresh empty baseline. Refresh preserves the authenticated Student session. If the baseline is occupied, record a setup blocker and do not begin dependent cases.

UAT-19 — Advisor login (execute after UAT-05, before UAT-06)

Field — UAT-19	Planned specification
Actor / wireframe	Advisor; Advisor identity/navigation in panel 3; shared login form is outside the original wireframe
Objective	Verify Advisor login, identity and isolated persistent session.
Preconditions	UAT-05 complete; A Pending; separate Advisor profile/private session signed out; Student window stays open
Test data	[Advisor test account] / [Advisor test password]; invalid password [invalid Advisor test password]
Steps	1. In the isolated Advisor window open http://127.0.0.1:5002/login. 2. Enter [Advisor test account] in **Username**, [invalid Advisor test password] in **Password**, and click **Sign in**. 3. Check the login error, then enter [Advisor test account] / [Advisor test password] and click **Sign in**. 4. Inspect **Advisor workspace**, Home, Dr. Williams and the **Advisor Review** sidebar link. Click **Profile** and check Username supplied for the Advisor test account and Role **Advisor administrator**. 5. Click **Home** and refresh. Confirm the Advisor session remains active. Refresh Student **Home** in its separate window and confirm it still shows the Student identity. Continue with UAT-06 in the Advisor window.
Expected result	Invalid credentials show **Incorrect username or password**. Valid login opens Advisor Home at /app, with Advisor navigation and no Degree Progress or Plan Courses sidebar links. It does not automatically open the review queue. Refresh retains the Advisor session, and the separate Student session is unaffected. If Student identity changes, stop and correct profile isolation before making decisions.

UAT-20 — Browser-visible role and signed-out restrictions (execute last)

Field — UAT-20	Planned specification
Actor / wireframe	Student (Advisor checks use the isolated Advisor window in steps 4–6); Different Student navigation in panels 1–2 and Advisor navigation in panel 3; access-denied pages are supporting implementation
Objective	Verify browser role restrictions and signed-out access denial.
Preconditions	UAT-17 complete; both role sessions authenticated and isolated; A/B Cancelled, new C Pending with E3
Test data	Direct URLs /admin, /app?tab=plan-courses and /app?tab=degree-progress; no plan IDs or request tampering
Steps	1. In Student **Home**, confirm **Degree Progress** and **Plan Courses** links exist and **Advisor Review** is absent. 2. In that Student window type http://127.0.0.1:5002/admin directly into the address bar and load it. 3. Record the response. Return by typing http://127.0.0.1:5002/app?tab=home; confirm Student Home still opens. 4. In the Advisor window confirm **Advisor Review** exists while **Degree Progress** and **Plan Courses** sidebar links are absent. 5. In the Advisor address bar load http://127.0.0.1:5002/app?tab=plan-courses, then http://127.0.0.1:5002/app?tab=degree-progress. Inspect the content after each load. 6. Click **Advisor Review** and verify the Student/C Pending card still contains only ENGL 1000, 3 credits. Make no decision. 7. In the Student window click **Sign out**. Type http://127.0.0.1:5002/app in the address bar and load it afresh (do not rely on a cached Back page). Inspect the response, then click **Return to Login**. 8. In the Advisor window click **Sign out**, then directly load http://127.0.0.1:5002/admin. Inspect the response and click **Return to Login**.
Expected result	Student direct access to /admin is denied with **You are not authorized to access this resource** (HTTP 403; the browser can display JSON, not a custom denial dashboard). No queue or review controls are exposed. Advisor attempts to use Student-only tab query values render **Advisor Home** at /app, because unsupported tabs fall back to Home; they do not display a Student planning/progress screen and do not promise HTTP 403. The valid Advisor queue remains accessible with unchanged C. After sign-out, direct /app and /admin loads show the implemented **401 Unauthorized / Session Expired** page with **Return to Login**. No plan state changes occur. This case demonstrates these specific browser restrictions only. It does not test forged POSTs, all API permissions, or another Student's ownership. Do not mark those as manually passed by inference.

Source: docs/assignment2/test-plans/uat-test-plan.md; reports/uat-execution.md (planned fields only).

6.2 Unit Test Plan

Objective. Verify business decisions in isolation from HTTP, templates and persistence. Test the real AdvisingService with autospecced repository doubles and in-memory SQLModel objects; reject accidental database connections. AuthService orchestration uses mocked repository/security collaborators. UT-23 exercises real local cryptography with controlled time and test-only settings, without database or network access.

Components. AdvisingService dashboard totals, prerequisite eligibility, semester/selection validation, submission delegation, plan edit/review/cancel/removal rules; AuthService; current-user/role dependencies; password/JWT utilities. UT-01–23 identify specifications, not counts of parameterized executions.

6.2.1 F1 fixture and test design

Fixture / design	Planned specification
Actors / terms	S1 Student id 1; S2 independent Student; A1 Advisor. T1=Year 1 semester 1; T2=Year 1 semester 2.
Courses	C1 id101 COMP 1100, Core; C2 id102 MATH 1000, General Education; C3 id103 COMP 2000, Core, requires C1; C4 id104 MATH 2000, Electives, no prerequisite; C5 id105 COMP 3000, requires C1 and unknown MATH 9999. Each is 3 credits.
Default state	C1 Completed; C2 In Progress; C3/C4 untracked; 93-credit degree; Core requirement 30; empty plan history. Cases override explicitly. F1 is synthetic test data, distinct from the browser seed.
Positive / negative / boundary	Positive: eligible selections and legal decisions. Negative: invalid terms, duplicates, missing/foreign resources, bad identity. Boundary: zero totals, cap at 100%, duplicate records, last remaining course and 0/499/500/501 raw feedback characters.
Oracle	Literal independently expected values and exact repository calls/no calls. Never mock the method under test or derive the expected answer by calling it. No observed outcomes are included in the tables below.

Case / planned inputs and setup	Expected output (not an observed result)
UT-01 — Degree totals and category calculation **Component:** AdvisingService.get_student_dashboard **Preconditions:** Mock repositories return fixture S1 and no plans. **Inputs:** F1 baseline: completed C1=3; in-progress C2=3; program=93; Core requirement=30.	Completed=3; in-progress=3; remaining=87; total=93; progress=3%; Core earned=3, remaining=27, percent=10. E01 already checks baseline persistence: add only isolated calculation assertions.
UT-02 — Empty academic record and missing program fallback **Component:** AdvisingService.get_student_dashboard **Preconditions:** Mock user/program/record lookups; no plans. **Inputs:** Variants: existing 93-credit program with no records; missing degree program.	Credits=0/0, remaining=93, progress=0; category earned=0 for configured requirements; absent program uses 93 and category_progress=[].
UT-03 — Progress limits, zero denominators and duplicate records **Component:** AdvisingService.get_student_dashboard **Preconditions:** Repository doubles only; deliberately synthetic boundary data. **Inputs:** Variants: total=6, completed=9, in-progress=3; total=0; category required=0; repeated completed record for C1.	Remaining never negative; progress and category percent capped at 100; zero denominator yields 0; duplicate same-status course records count its credits once. Do not claim service sanitizes all inconsistent records.

Case / planned inputs and setup	Expected output (not an observed result)
UT-04 — Prerequisites satisfied or absent **Component:** AdvisingService.prerequisite_ok; get_course_availability **Preconditions:** Mock catalogue, completed IDs and course records. **Inputs:** C3 requires C1; C4 requires none; advanced course requires ' COMP 1100, MATH 1000 '; both completed in last variant.	Untracked targets eligible if all named prerequisites completed; whitespace ignored; no prerequisites eligible; availability=(True, 'Eligible').
UT-05 — Missing and in-progress prerequisites **Component:** AdvisingService.prerequisite_ok; get_course_availability **Preconditions:** Mock completed IDs exclude C2; C2 is In Progress. **Inputs:** C3 requiring C2; advanced course requiring C1 and unknown MATH 9999; blank comma-separated tokens.	In-progress prerequisite does not satisfy completion; unknown code fails; availability names only unmet codes; empty comma tokens ignored. E08 supplies DB-backed missing-code baseline.
UT-06 — Tracked or unknown target courses **Component:** AdvisingService.prerequisite_ok; get_course_availability **Preconditions:** Mock repository returns completed C1, in-progress C2 and absent ID 999999. **Inputs:** Targets C1, C2, 999999.	False for all three; C1 says Already completed; C2 says Already in progress; nonexistent course absent from availability map.
UT-07 — Valid semester names and trimming **Component:** AdvisingService.submit_plan; SEMESTER_OPTIONS **Preconditions:** Mock eligible C3, no existing plan, create_plan sentinel. **Inputs:** All nine Year {1..3} semester {1..3}; padded ' Year 1 semester 1 '.	Each accepted; exactly nine options including semester 3; padded input stored trimmed. E06 baseline reused rather than copied.
UT-08 — Invalid semester names **Component:** AdvisingService.submit_plan **Preconditions:** Mock repositories; mutation methods are spies. **Inputs:** '', whitespace, Fall 2026, Year 0 semester 1, Year 4 semester 1, Year 1 semester 4, Year 1 Semester 1.	ValueError 'Choose a valid year and semester before submitting a plan.'; no create/resubmit call. E07 reused for its legacy-name variant.
UT-09 — Empty selection **Component:** AdvisingService.submit_plan **Preconditions:** No database; repository spies. **Inputs:** [] with valid T1.	ValueError 'Select at least one course for the semester plan.'; no persistence or eligibility lookup.
UT-10 — Duplicate selected course IDs **Component:** AdvisingService.submit_plan **Preconditions:** No database; repository spies. **Inputs:** [C3,C3] and [C3,C4,C3] with T1.	ValueError 'A course may only appear once in a semester plan.'; no create/resubmit call. Distinct from duplicate semester plan in E05.
UT-11 — Missing or ineligible selection aborts submission **Component:** AdvisingService.submit_plan **Preconditions:** Mock availability and course lookup, no real repository operations. **Inputs:** [C3,999999]; [C1]; [C2]; [advanced course missing MATH 9999].	Missing ID reports no longer available; ineligible selection includes course code and reason; no create/resubmit call even when earlier list item was valid.

Case / planned inputs and setup	Expected output (not an observed result)
UT-12 — Submission status decision table **Component:** AdvisingService.submit_plan **Preconditions:** Mock existing-plan lookup and mutation return sentinels. **Inputs:** Existing status: none, Pending, Approved, Rejected, Changes Requested, Cancelled; valid [C3],T1.	None/Cancelled -> create_plan; Rejected/Changes Requested -> resubmit_plan(existing,[C3]); Pending/Approved -> ValueError already under review or approved and no mutation. Database replacement/reset details belong to integration tests.
UT-13 — Dashboard edit and cancelled-selection rules **Component:** AdvisingService.get_student_dashboard **Preconditions:** Mock current plan and its C3 link, no DB. **Inputs:** Each status Pending, Approved, Rejected, Changes Requested, Cancelled; also no plan.	Edit true only Rejected/Changes Requested/Cancelled; false Pending/Approved/no plan. Cancelled hides saved course links; other existing statuses retain selected C3.
UT-14 — Advisor decisions and feedback normalization **Component:** AdvisingService.review_plan **Preconditions:** Mock Pending plan, update_plan sentinel. **Inputs:** approve, reject, request_changes; feedback=' Review note '; advisor=A1.	Mapped statuses Approved, Rejected, Changes Requested; feedback='Review note'; advisor_id=A1; exactly one update call. Empty feedback is allowed for every decision.
UT-15 — Invalid advisor decision **Component:** AdvisingService.review_plan **Preconditions:** Repository lookup/update spies. **Inputs:** '', APPROVE, cancel, unknown; valid short feedback.	ValueError 'Choose approve, reject, or request changes.' before persistence; no update call.
UT-16 — Feedback length boundary before trimming **Component:** AdvisingService.review_plan **Preconditions:** Mock Pending plan and update spy. **Inputs:** For each legal decision: lengths 0,499,500,501; 501 spaces; ' '+499 characters+' ' (501 total).	0/499/500 accepted, stored stripped; 501 rejected with 'Advisor feedback must be 500 characters or fewer.' even if trimming would shorten it; no update on rejection.
UT-17 — Review only existing Pending plans **Component:** AdvisingService.review_plan **Preconditions:** Mock lookup absent or returning status variant. **Inputs:** Missing plan; Approved, Rejected, Changes Requested, Cancelled; valid approve/feedback.	Missing -> ValueError 'Plan not found'; each non-Pending -> 'Only pending plans can be reviewed.'; no update.
UT-18 — Cancellation status and ownership matrix **Component:** AdvisingService.cancel_plan **Preconditions:** Mock owned/foreign/missing plan and cancel spy. **Inputs:** Owned Pending, Approved, Rejected, Changes Requested, Cancelled; foreign S2 plan; missing plan.	Only owned Pending/Approved/Rejected delegated to repo.cancel_plan; other owned statuses reject; foreign/missing -> 'Plan not found.'; no mutation on denial.
UT-19 — Remove course guard matrix **Component:** AdvisingService.remove_submitted_course **Preconditions:** Mock owned Pending plan with C3/C4 links and remove spy. **Inputs:** Remove C4 from two links; last C3; absent course; non-Pending statuses; foreign or absent plan.	Only an owned Pending plan with matching course and more than one link permits deletion; others reject with existing guard messages; no deletion on failure.

Case / planned inputs and setup	Expected output (not an observed result)
UT-20 — Authentication success **Component:** AuthService.authenticate_user **Preconditions:** Mock UserRepository; patch verify_password and create_access_token in app.services.auth_service. **Inputs:** Stored user id=S1, role=regular_user, password='stored-hash'; verifier=True.	Returns token sentinel; verifier receives plaintext and stored hash; token data={'sub': str(S1), 'role':'regular_user'}; no real DB/hash/JWT needed.
UT-21 — Authentication failures and short circuit **Component:** AuthService.authenticate_user **Preconditions:** Mock repository and security collaborators. **Inputs:** Unknown username; known user with verifier=False.	Returns None; missing user never invokes verifier; neither case issues token. No token or password material recorded as evidence.
UT-22 — Cookie token resolution and admin role **Component:** get_current_user; is_admin_dep in app.dependencies.auth **Preconditions:** Synthetic Request, mocked jwt.decode/settings/UserRepository; async invocation. **Inputs:** Missing cookie; decoder InvalidTokenError; nonexistent subject user; valid S1; valid A1.	Missing/invalid/unresolved identity raises HTTP 401; valid identity resolves the repository user. Current require_roles denies regular_user at the Advisor guard with 403 and returns admin. Test the current database role, independently of a token role claim.
UT-23 — Password hash and JWT utility contracts **Component:** encrypt_password; verify_password; create_access_token **Preconditions:** No DB; actual local hash/JWT utilities, fixed test settings and controlled clock. **Inputs:** Test-only password; correct/wrong password; claims sub=S1,role=regular_user; explicit 5-minute lifetime.	Hash differs from plaintext; correct verifies true, wrong false; signed token carries sub/role and exp at fixed time +5 minutes; input dict not mutated. Patch time/settings only, not functions under test.

Current role-contract clarification for UT-22 is adapted from the actual require_roles implementation (401 identity / 403 wrong role). Other original inputs/oracles are retained. E01/E06/E07/E08 references denote existing database-backed baseline tests in tests/test_advising_service.py; their original planning notes avoid duplicate coverage.

Source: docs/assignment2/test-plans/unit-test-plan.md; tests/unit/test_advising.py; tests/unit/test_auth.py.

6.3 Integration Test Plan

Objective and boundary. Join real FastAPI routes/dependencies, form/JSON parsing, real cookie/JWT authentication, services, repositories and SQLModel persistence. Exercise Student/Advisor workflows and database rollback. IT-19/20 deliberately reach database failures directly because service validation rejects those inputs before SQL.

Setup / coverage	Planned specification
F2 data	Reuse F1 course/degree meanings but resolve generated row IDs. Seed two Students S1/S2 and Advisor A1, real password hashes, C1 Completed/C2 In Progress for S1, distinctive S2-only records/feedback, and independent plan state per variant.
Isolation	Fresh per-case SQLite engine (StaticPool, check_same_thread=False), metadata and foreign keys enabled. New Session per HTTP request and per persistence check. Separate Student/Advisor cookie jars; follow_redirects=False.
Startup and cleanup	Rebind test engine/session dependencies and startup/recovery initialization so normal/private DBs cannot be touched. Do not replace authentication, handlers, business methods or CRUD. Close clients/sessions, restore overrides/lifespan/settings and dispose engine.
Coverage	Real login/logout; invalid credentials/tokens; current DB roles; course eligibility; Student submission and persistence; Advisor queue and decisions; feedback/resubmission; removal/cancellation/replacement; ownership/privacy; invalid schemas, constraints and atomic rollback.
Current contracts	Unauthenticated protected HTML/JSON: 401; authenticated wrong-role guard: 403. HTML business errors: 303 + warning flash. Strict JSON APIs use documented 401/403/404/409/422 semantics. Historical H3/HTTP-200 plan wording is obsolete and adapted below.
Limits	SQLite and local HTTP do not prove PostgreSQL, production cookies/CSRF, concurrency or human usability. Current runtime-engine and corrected browser-style tests supplement, rather than replace, F2.

Case / planned input and procedure	Expected output (not an observed result)
IT-01 — Real login, cookie and logout **Component:** POST /login; GET /app; GET /logout **Preconditions:** F2 database, separate unauthenticated client, real hash/JWT/auth dependencies. **Inputs:** S1 correct password; wrong password; unknown username. **Procedure:** POST form credentials for each variant; capture cookie/redirect; with successful client GET /app; GET /logout then protected GET.	Success ->303 /app and access_token cookie, authenticated page 200; wrong/unknown ->303 /login, Incorrect username or password flash, no new access_token on clean client; logout ->303 /login, cookie removed, protected HTML/JSON access denied with HTTP 401.
IT-02 — Student degree progress through HTTP **Component:** GET /app?tab=degree-progress&semester=T1 **Preconditions:** F2; login S1 using IT-01 setup. **Inputs:** C1 Completed=3; C2 In Progress=3; 93-credit program; Core required=30; S2 distinct records. **Procedure:** GET page; inspect rendered totals and category; query records in fresh session.	200; completed=3, in-progress=3, remaining=87, overall=3%; Core earned=3/30 and remaining=27; no S2 activity or profile; no database mutation.

Case / planned input and procedure	Expected output (not an observed result)
IT-03 — Eligible selection, submission and advisor queue **Component:** GET /app?tab=plan-courses; POST /app/plan; GET /admin **Preconditions:** F2; S1 and A1 authenticated in separate clients; no T1 plan. **Inputs:** Repeated form course_ids=C3 and C4; semester_name=T1. **Procedure:** GET catalogue; inspect eligible/disabled options; POST form selection; follow redirect manually; GET /admin as A1; query saved plan/links.	Catalogue eligibility reflects completed C1 and in-progress C2; POST ->303 with semester and tab=plan-courses; success flash; exactly one Pending S1/T1 plan, two unique links, submitted_at set, advisor_id=None; appears in real advisor queue.
IT-04 — Persistence across requests and sessions **Component:** POST /app/plan; GET /app; AdvisingRepository.get_plan_for_student/get_plan_courses **Preconditions:** F2; S1 authenticated; no T1 plan. **Inputs:** [C3,C4],T1. **Procedure:** Submit through HTTP; close request-scoped session; open a fresh Session on same isolated engine; read plan/links; issue another GET.	Committed plan ID, S1, T1, Pending and both links survive session boundaries and render as selected; no reliance on Python object identity. E02/E09 do not prove full authenticated request chain.
IT-05 — Advisor approval and rejection **Component:** POST /admin/review/{plan_id}; GET /admin; GET /app?tab=messages **Preconditions:** F2; seed a separate Pending S1 plan for each decision; S1 and A1 logged in. **Inputs:** approve with ''; reject with 'Please choose another course.'; JSON and form variants. **Procedure:** A1 POST each review; inspect redirect/flash; read row in fresh session; GET advisor queue and student messages.	303 /admin; matching success flash; Approved or Rejected stored with A1 and stripped feedback; reviewed plan leaves Pending queue; student sees own decision/message; plan course links preserved.
IT-06 — Request changes and resubmit **Component:** POST /admin/review/{id}; POST /app/plan; GET /app and /admin **Preconditions:** F2; Pending S1/T1 plan containing C3, A1/S1 authenticated. **Inputs:** request_changes, feedback=' Choose another course. '; replacement=[C4]. **Procedure:** A1 request changes through HTTP; verify persisted status/student message; S1 submit replacement for same semester; query fresh session and advisor queue.	Changes Requested, A1, trimmed feedback initially; resubmission ->303; same plan ID becomes Pending; links exactly [C4]; old C3 link removed; advisor_id=None, feedback=''; submitted_at refreshed; returns to Pending queue.
IT-07 — Rejected plan resubmission without cancellation **Component:** POST /admin/review/{id}; POST /app/plan **Preconditions:** F2; S1/A1 authenticated; S1 Pending T1 plan containing C3. **Inputs:** reject with feedback; replacement=[C4]. **Procedure:** Reject via HTTP; submit C4 for same semester without cancel; read fresh-session plan and links.	303 for both requests; same ID reset to Pending; C4 replaces C3; feedback and advisor cleared; timestamp refreshed. Distinct from E10 cancellation/new-ID behavior.

Case / planned input and procedure	Expected output (not an observed result)
IT-08 — Allowed cancellation and replacement **Component:** POST /app/plan/{id}/cancel; POST /app/plan; GET /admin and /app **Preconditions:** F2; independently owned Pending, Approved and Rejected plans; S1 logged in. **Inputs:** Each allowed status with existing [C3] links; replacement [C4], same semester. **Procedure:** Cancel per variant; inspect row, links, UI and advisor queue; submit replacement; query all matching plan rows.	303; original row Cancelled and retains links/history; hidden from Pending queue; cancelled plan displays no selected/planned courses; replacement has new ID and Pending status, old Cancelled row remains.
IT-09 — Forbidden cancellation statuses **Component:** POST /app/plan/{id}/cancel **Preconditions:** F2; S1 authenticated; independently owned Changes Requested and Cancelled plans. **Inputs:** Each disallowed status; unchanged links/feedback snapshots. **Procedure:** POST cancel per variant; follow redirect for warning; query fresh state.	303 tab=plan-courses with 'Only pending, rejected, or approved plans can be cancelled.' warning; status, links and feedback unchanged. Changes Requested permits resubmission, not cancellation.
IT-10 — Submitted course removal and last-course rule **Component:** POST /app/plan/{id}/courses/{course_id}/remove **Preconditions:** F2; S1 authenticated, owned Pending plan [C3,C4]; separate non-Pending variants. **Inputs:** C4 then C3; absent course ID; each Approved/Rejected/Changes Requested/Cancelled. **Procedure:** Remove C4; try last C3; test absent and non-Pending variants independently; query DB after each.	First request 303/success, one C3 link remains; last-course, absent-course and non-Pending requests 303/warning without state change; no orphaned/deleted plan.
IT-11 — Cross-student ownership and data visibility **Component:** GET /app?tab=degree-progress; GET /app?tab=messages; POST /app/plan/{id}/cancel; POST /app/plan/{id}/courses/{course_id}/remove; POST /app/plan **Preconditions:** F2; S1 and S2 logged in separately; S2 owns Pending [C3,C4] and distinctive records/feedback. **Inputs:** S1 uses S2 plan ID for cancel/remove; submits valid own selection with extra student_id=S2 field. **Procedure:** S1 GET dashboard/messages; POST foreign cancel/remove; POST own plan with forged student_id field; compare DB before/after.	Foreign operations ->303 warning 'Plan not found.' with S2 row/links untouched; no S2 private record/message in S1 output; new submission belongs to cookie-authenticated S1, extra form student_id does not override identity.
IT-12 — Unauthenticated and invalid-token access **Component:** GET /app; GET /admin; POST /app/plan; POST /admin/review/{id}; POST /app/plan/{id}/cancel; POST /app/plan/{id}/courses/{course_id}/remove **Preconditions:** F2; valid target plan; clean client per token variant; no auth override. **Inputs:** No cookie; malformed token; expired correctly signed token; wrong-signature token; valid token for nonexistent user. **Procedure:** Send each protected request with otherwise valid payload under each variant; compare entire relevant DB state.	Protected HTML and JSON requests return HTTP 401 for missing/malformed/expired/wrong-signature/nonexistent-user tokens, with no private data and no database writes. This replaces the obsolete H3/HTTP-200 expectation.

Case / planned input and procedure	Expected output (not an observed result)
IT-13 — Student cannot use advisor operations **Component:** GET /admin; POST /admin/review/{id} **Preconditions:** F2; S1 logged in; Pending S2 plan. **Inputs:** Valid approve/reject/request_changes payloads as regular_user. **Procedure:** GET queue; POST each decision; query plan in fresh session.	Authenticated Student access to Advisor queue/review returns HTTP 403. No queue/private data or status/feedback/advisor/course-link changes. This replaces obsolete H3/HTTP-200 wording.
IT-14 — Business-invalid submission inputs **Component:** POST /app/plan **Preconditions:** F2; S1 logged in; no plan for target semester; reset DB per variant. **Inputs:** Omitted/empty course_ids; [C3,C3]; [999999]; [C1] completed; [C2] in progress; unmet prerequisite C5; invalid/blank semester; valid C3 plus invalid item. **Procedure:** POST form per variant; capture redirect; GET warning; compare SemesterPlan/PlanCourse counts.	303 warning for each business rejection, no partial plan/links. Messages identify empty/duplicate/missing/ineligible selections or invalid semester. Whitespace around a valid semester is accepted and stored trimmed (positive control).
IT-15 — Duplicate active semester plan **Component:** POST /app/plan **Preconditions:** F2; S1 logged in; owned T1 plan separately Pending or Approved. **Inputs:** Same-semester [C4]; second positive control uses T2. **Procedure:** POST duplicate; snapshot row/links; then submit to T2.	Pending/Approved duplicate ->303 warning already under review or approved; original plan/links unchanged, no extra T1 plan; independent T2 submission accepted.
IT-16 — Invalid review payload shapes and syntax **Component:** POST /admin/review/{id} **Preconditions:** F2; A1 logged in, Pending plan; reset per variant. **Inputs:** Malformed JSON '{'; JSON [] or null; {}; decision='' or integer; feedback=null/list/integer; noninteger plan path ID; valid form control. **Procedure:** POST with accurate Content-Type per variant; capture raw response; compare persisted snapshot.	Malformed JSON ->400; non-object/missing-or-invalid decision/non-string feedback/noninteger path ->422; no state changes; valid form decision with string feedback ->303 and normal review. JSON syntax checks must send raw malformed bytes, not client JSON serialization.
IT-17 — Review business validation and feedback boundary **Component:** POST /admin/review/{id} **Preconditions:** F2; A1 authenticated; independent Pending/non-Pending/missing plan cases. **Inputs:** Unknown decision; feedback lengths 0,499,500,501; 501 spaces; non-Pending statuses; missing integer ID. **Procedure:** POST JSON and form boundary variants; follow redirect for flash; query fresh state.	Business-invalid cases ->303 /admin warning, unchanged DB; valid 0/499/500 feedback accepted and trimmed; 501 rejected before trimming. Missing plan says Plan not found; non-Pending says Only pending plans can be reviewed. No invented mandatory-feedback rule.
IT-18 — Invalid form types and path identifiers **Component:** POST /app/plan; POST /app/plan/{id}/cancel; POST /app/plan/{id}/courses/{course_id}/remove **Preconditions:** F2; S1 logged in, owned Pending [C3,C4]. **Inputs:** course_ids='abc' or mixed [C3,'abc']; plan_id='abc'; course_id='abc'; missing integer plan ID 999999; missing course membership. **Procedure:** Send repeated form fields or path variants; capture validation response and DB snapshot.	Noninteger form/path IDs ->422 before service mutation; nonexistent integer plan or absent plan course ->303 warning; plan and links unchanged. Student submission endpoint expects form data; do not assume a JSON REST contract.

Case / planned input and procedure	Expected output (not an observed result)
IT-19 — Database uniqueness constraints **Component:** PlanCourse and StudentCourse SQLModel tables; AdvisingRepository.create_plan **Preconditions:** F2 fresh engine with foreign keys enabled; seed referenced users/courses and valid Pending plan. **Inputs:** Duplicate (plan_id,C3); duplicate (S1,C1); invalid foreign-key link. **Procedure:** Use isolated repository/session boundary to insert each invalid row; catch IntegrityError; rollback; open fresh session.	Real database rejects duplicate pairs and dangling foreign key; committed valid rows remain intact. This is a DB integration case rather than HTTP; UT-10 covers the earlier service guard.
IT-20 — Create/resubmit transaction rollback **Component:** AdvisingRepository.create_plan; resubmit_plan **Preconditions:** F2 real isolated DB; existing Rejected plan [C3] for resubmit case. **Inputs:** Create with [C4,C4]; resubmit existing plan with [C4,C4] to trigger real unique-constraint failure. **Procedure:** Invoke repository directly to reach DB failure past service duplicate guard; catch IntegrityError; inspect a fresh session after rollback.	Failed create leaves no new plan or links; failed resubmit preserves original status/feedback/advisor/timestamp/C3 links; session usable after rollback. Keep actual DB/CRUD real; do not mock commit. Normal route persistence remains covered by IT-03/04/06.

IT-01/12/13 explicitly use the current 401/403 contract in place of historical H3. The original source documents remain attached unchanged. These adaptations are planned expectations, not new observations.

6.3.1 Corrected-build supplemental plan

The retained correction plan adds HTML Student-role denials, actual application SQLite engine enforcement, malformed registration email and computed table/Approved-notice contrast. Expect 403 and unchanged full database snapshots for admin/unknown-role HTML mutations; 422 and no registration writes for malformed email; IntegrityError for orphan links on separately pooled application-engine connections; and readable contrast in normal/hover/Approved states. Manual browser and full Postman retests are still planned, not executed results.

Source: docs/assignment2/test-plans/integration-test-plan.md; reports/correction-retest-guide.md; tests/integration/test_audit_corrections.py; test_contrast_regression.py.

7. Implementation

7.1 GitHub Repository

Repository: https://github.com/Camcoon/MyAdvisorWithTests. Default branch **main**. Live read-only GitHub verification on 27 October 2026 found head 864386fe10fa5e61869e07500d84e51ff7e46b9b; its complete recursive tree contains 430 entries and is not truncated.

Publication status (latest check): the intended MyAdvisorWithTests repository contains an earlier Postman Collection JSON export, but its latest revised PM19/PM20 assertions and the full corrected local code/evidence have not been verified as published. The agent prepared a separate clean upload checkout and reports that no commit/push occurred. This PDF does not publish code or files to GitHub; the user will update the repository manually.

Artifact	Repository / local verification
Unit implementation	tests/unit/test_advising.py and tests/unit/test_auth.py are present on main.
Integration implementation	tests/integration/test_student_workflow.py, tests/integration/test_advisor_workflow.py and tests/integration/test_database_integrity.py are present. Correction tests and four changed application/CSS files remain local.
Postman JSON	Local 90-request Collection v2.1 JSON was validated. An earlier export is now present on GitHub; the newer version with strengthened requests 19/20 has not been verified as published. See Section 7.4 and Appendix B.
Execution evidence	Six unit/integration PNGs are published. Runner/UAT images and corrected-build text/XML/hash evidence are retained locally; relevant evidence is embedded in this PDF.

7.2 Unit Test Implementation

pytest markers select unit tests with -m unit. tests/unit/conftest.py patches repository constructors where AdvisingService imports them, supplies autospecced doubles/in-memory model objects, and guards database connections. It preserves the real service logic and tracks mutation calls. test_advising.py implements UT-01–19; test_auth.py implements UT-20–23. An existing mocked login companion is also in the historical selection.

The functions under test include get_student_dashboard, prerequisite_ok, get_course_availability, submit_plan, review_plan, cancel_plan, remove_submitted_course, authenticate_user, get_current_user, the role guard and password/JWT utilities. Parameterization covers legal/illegal status matrices and raw feedback-length boundaries. UT-23 uses real hashing/signing; credentials and tokens are never printed here.

7.2.1 Actual unit implementation example

```
def test_UT_09_empty_selection(advising):
    with pytest.raises(ValueError, match="^Select at least one course for the semester plan\\.$"):
        advising.service.submit_plan(1, [], T1)
    no_submission(advising)
    advising.repo.list_courses.assert_not_called()
    advising.repo.get_completed_course_ids.assert_not_called()
    advising.repo.get_completed_courses.assert_not_called()
```

Verbatim test from tests/unit/test_advising.py. no_submission asserts that neither create_plan nor resubmit_plan was called. This checks both the error and early short-circuit behavior; Section 6 contains the plan and Section 8 contains observed execution.

7.3 Integration Test Implementation

FastAPI/Starlette TestClient sends real requests through the application. tests/integration/conftest.py creates a fresh SQLite engine and schema per case, enables foreign keys, isolates startup/recovery, overrides request sessions and opens independent actor clients. Real password hashing, JWT decoding, current database role lookup, request parsing, services and CRUD remain active. Teardown restores app settings/overrides and closes database resources.

Test file	Actual implementation
tests/integration/test_student_workflow.py	Student login/cookies/logout, progress/catalogue, form and JSON submission, committed state, ownership/privacy, validation, removal, cancellation and same-ID resubmission.
tests/integration/test_advisor_workflow.py	Real Student-to-Advisor queue delivery; decisions through form, generic JSON and named JSON actions; feedback; resubmission and replacement; authorization and validation boundaries.
tests/integration/test_database_integrity.py	Real uniqueness/foreign-key constraints and rollback for failed create/resubmit; fresh-session checks of committed state.
tests/integration/test_audit_corrections.py	18 additional variants: 12 HTML role denials, 3 malformed-email no-write cases, valid/duplicate registration, and 2 actual engine-integrity cases.
tests/integration/test_contrast_regression.py	2 additional variants use the rendered templates and pinned CSS in Chromium; normal/hover table and Approved explanatory text contrast. Browser harness is tests/browser/contrast_probe.cjs.

7.3.1 Actual workflow implementation example

```
def test_cancelled_plan_leaves_advisor_queue_and_replacement_returns(student):
    env, client = student.env, student.client
    advisor = env.login("a1")
    plan_id = env.seed_plan()
    queue(advisor, [plan_id])
    before = env.snapshot()
    response = client.post(f"/app/plan/{plan_id}/cancel")
    assert response.status_code == 303
    assert "Semester plan cancelled." in client.get(response.headers["location"]).text
    assert env.plan(plan_id) == {**before["semesterplan"][0], "status": "Cancelled"}
    assert env.snapshot()["plancourse"] == before["plancourse"]
    queue(advisor, [])
```

Verbatim opening of the cancellation/queue/replacement test. A real Student cancels a Pending plan, a fresh database snapshot verifies Cancelled status and retained links, and the real Advisor queue becomes empty. The remainder submits a replacement and verifies the new Pending plan. No browser UAT conclusion is inferred.

Separate S1/S2/A1 clients check missing/invalid/expired tokens, current roles, Student denial from Advisor operations and concealment of foreign plans. Negative tests compare committed snapshots after denied writes. tests/test_advising_api.py, test_semester_plans_api.py, test_advisor_plans_api.py and test_api_security.py provide additional API/regression coverage in the full suite.

7.4 API Test Implementation — Postman

Collection: MyAdvisor - COMP 3613 Assignment 2. **Format:** Postman Collection v2.1.0 (schema format, not a release number). **Filename/path:** docs/assignment2/postman/MyAdvisor-Assignment2.postman_collection.json. SHA-256: 409cffa7540cd73880d74139f77346cdb47d06a18d9ae8e1b11cf70ef2c44b6d.

GitHub JSON (earlier export confirmed): https://github.com/Camcoon/MyAdvisorWithTests/blob/main/docs/assignment2/postman/MyAdvisor-Assignment2.postman_collection.json
Revised local JSON strengthens requests 19/20 and passed static validation. Its publication and Runner execution are unverified. The prior Postman results remain historical.

Ordered request group	Requests	Coverage
01 Readiness and real cookie authentication	01–12 (12)	Health; anonymous access; malformed cookie; bad/valid Student and Advisor login; independent Student registration/login.
02 Role boundaries, progress and catalogue	13–21 (9)	Role denials; response privacy; progress; dynamic eligible/blocked course discovery; free-semester selection.
03 Student plan lifecycle and validation	22–50 (29)	Empty/duplicate/unknown/ineligible/invalid submissions; persistence; foreign ownership; duplicate active plan; Pending removal and last-course protection.
04 Advisor decisions and resubmission	51–76 (26)	Queue/detail, invalid decisions/shapes, 500/501-character feedback, change request, same-ID resubmission, reject and approve.
05 Cancellation, history and logout	77–90 (14)	Preserved approval, cancellation/history/new-ID replacement, queue cleanup, actor logout and no-cookie denial.

Authentication/cookies. Login captures each actor’s access_token into separate collection variables. Requests send the intended actor’s Cookie explicitly; the prescribed run disables automatic cookie-jar and redirect handling. HttpOnly cookies are server controls; exported password/token values are blank. Populate run-only credentials privately on a disposable instance. Do not use a normal/private database.

Dynamic state. Collection variables include base_url, actor tokens, discovered course IDs, selected semester and generated plan IDs. Scripts validate response status/schema/privacy, select eligible data, assert persisted decisions and read back after denials. Request 01 initializes run state. Execute one iteration in the original 01–90 order; later requests depend on earlier state.

Positive/negative coverage. Valid operations are paired with expected 401/403/404/409/422 denials, malformed JSON/IDs, spoofed fields, foreign resources, illegal status transitions and feedback boundaries. Static validation parses 90 requests, compiles 93 JavaScript script blocks and covers all 15 JSON operations; static success is distinct from Runner execution.

Source: docs/assignment2/postman/MyAdvisor-Assignment2.postman_collection.json; postman/validate_collection.cjs; docs/assignment2/api.md.

7.4.1 API endpoint coverage

Method	Implemented JSON endpoint
GET	/api/degree-progress
GET	/api/courses
GET / POST	/api/semester-plans
GET / DELETE	/api/semester-plans/{plan_id}
PUT	/api/semester-plans/{plan_id}/courses
DELETE	/api/semester-plans/{plan_id}/courses/{course_id}
GET	/api/advisor/plans
GET	/api/advisor/plans/{plan_id}
POST	/api/advisor/plans/{plan_id}/review
POST	/api/advisor/plans/{plan_id}/approve
POST	/api/advisor/plans/{plan_id}/reject
POST	/api/advisor/plans/{plan_id}/request-changes
GET	/api/users

The combined GET/POST and GET/DELETE rows each represent two operations: **15 JSON operations** in total. GET /health, POST /login, POST /register and GET /logout provide supporting readiness/session operations. HTML mutation denials and corrected registration behavior require supplemental new-build requests; the historical collection run does not attest them.

7.4.2 Actual collection-script excerpt

49 - Cannot remove final course

```
try {
let assertionFailed = false;
function check(name, fn) {
  pm.test(name, function() {
    try { fn(); } catch (error) { assertionFailed = true; throw error; }
  });
}
check("HTTP 409", () => pm.response.to.have.status(409));
```

Contiguous verbatim opening excerpt. The complete exported script is in the attached JSON. F13 remains: request 19’s percentage range checks are weaker than an exact independent percentage oracle, and initial course discovery alone trusts returned eligibility. No assertion enhancement or new Runner pass is claimed.

8. Test Reports

Execution evidence is separated by build and date. Logs/screenshots support earlier 715- and 735-pass full runs, the original 113/194 marked selections, and the 90-request Postman iteration. After the approved credit changes, Codex reported 46 affected tests and 751 full-suite tests passing; the 751-run log and screenshot were not supplied with this PDF.

Execution	Verified actual result	Build / evidence qualification
Unit-marker run	113 passed; 602 deselected; 1 warning; 6.23s	26 Oct historical terminal screenshots; numeric exit code/build hash absent.
Integration-marker run	194 passed; 521 deselected; 1 warning; 30.52s	26 Oct historical terminal screenshots; numeric exit code/build hash absent.
Supporting full regression	715 passed; 1 warning; 76.25s; exit 0	27 Oct pre-correction audit text/XML and source hashes.
Corrected full regression	735 passed; 1 warning; 58.11s; exit 0	27 Oct corrected source, text/XML, 70 source-file hashes; zero failures/errors/skips.
Corrected targeted suite	284 passed; 1 warning; 86.42s; exit 0	27 Oct affected/correction coverage; overlaps complete suite.
Postman Runner	90 requests; 470 assertions passed; 0 assertion failures; 0 execution errors	27 Oct original Runner; 1 iteration; 10s 697ms; exact tested build/browser/DB unrecorded.
Manual UAT (supporting)	1 Pass / 0 Fail / 19 Not Run	27 Oct latest confirmed disposition; 18 partial, UAT-05 unverified; corrected-build manual run pending.

Counts are not additive across independent runs. Historical markers selected 113 unit and 194 integration tests from 715 items. The attached 735-pass XML belongs to an earlier corrected build (113 unit / 214 integration / 408 other). The subsequently reported 751 passes supersede that build result, but no 751 XML or new marker-run screenshot is included. Earlier screenshots are not presented as retests of the latest implementation.

8.1 Unit Test Report

Objective: demonstrate isolated service/authentication rules and repository delegation. **Environment:** Windows/win32; repository .venv; Python 3.14.0; pytest 9.1.1; pluggy 1.6.0; anyio 4.15.1; pytest-asyncio 1.4.0; pyproject.toml. The historical screenshots do not establish a complete dependency lock or exact source/build identifier.

Actual command (from the start capture; run from the repository root):

```
$base = Join-Path $env:LOCALAPPDATA "MyAdvisor-A2-Pytest"
.\.venv\Scripts\python.exe -m pytest tests -m unit -v --basetemp="$base\unit"
```

Observed summary: 715 collected, 113 selected, 602 deselected; **113 passed, 1 warning in 6.23s**. Genuine final-result Figure 8.3 follows. No numeric exit code is visible, so none is invented. The latest marked scrollback/XML is incomplete; earlier retained unit XML and the full audit records provide separate supporting evidence.

Warning and conclusion: StarletteDeprecationWarning reports deprecated httpx use with starlette.testclient and suggests httpx2. It is a dependency-maintenance warning, not a failed assertion. The selected run passed its tested cases; it does not establish full UAT, production security or a corrected-build marked rerun.

8.1 Unit terminal evidence — run capture 1

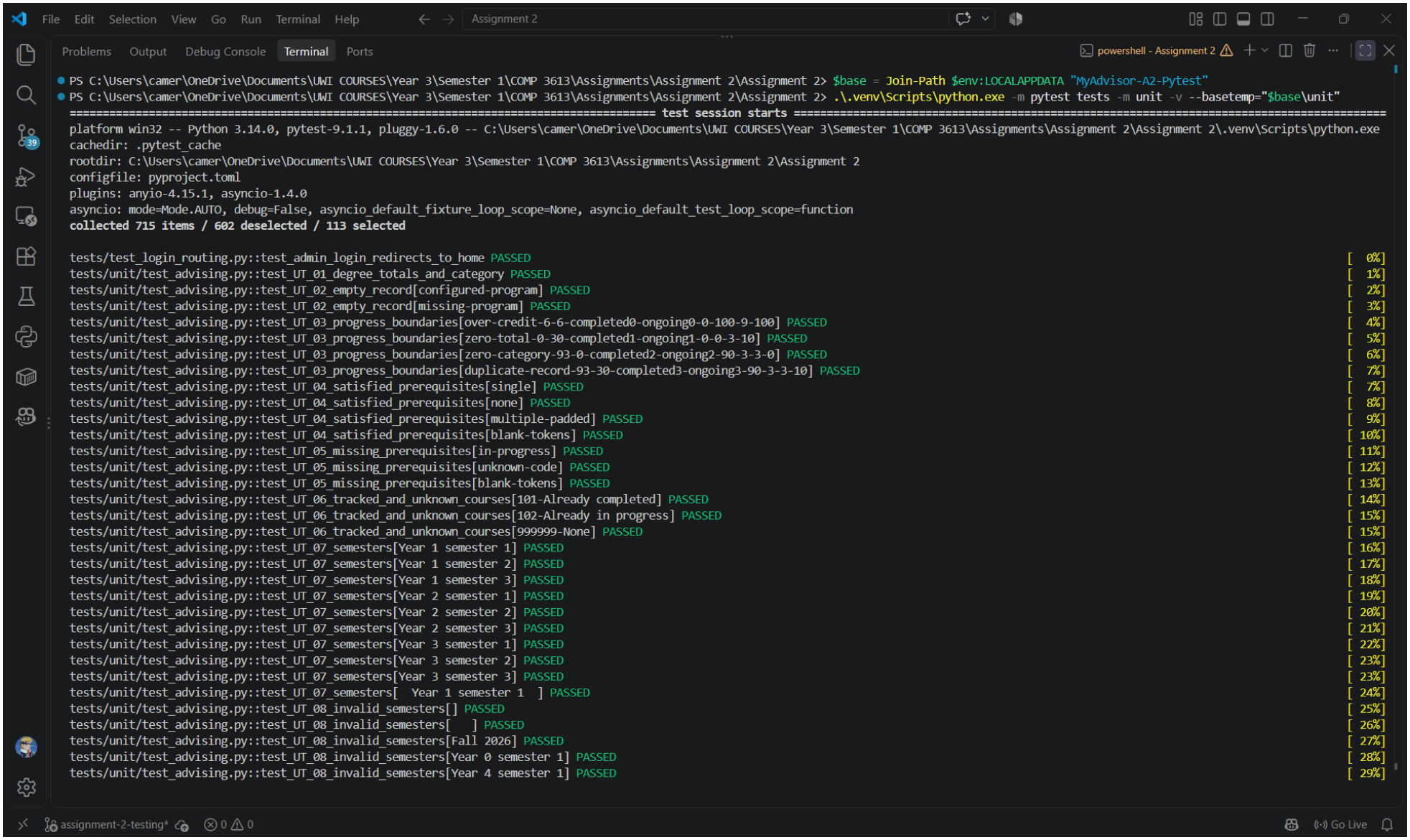


Figure 8.1. Genuine terminal capture Screenshot 2026-10-26 231703.png; 26 October 2026. Unmodified actual command/environment or passed variants; the final result is in Figure 8.3. Historical source attribution remains qualified.

8.1 Unit terminal evidence — run capture 2

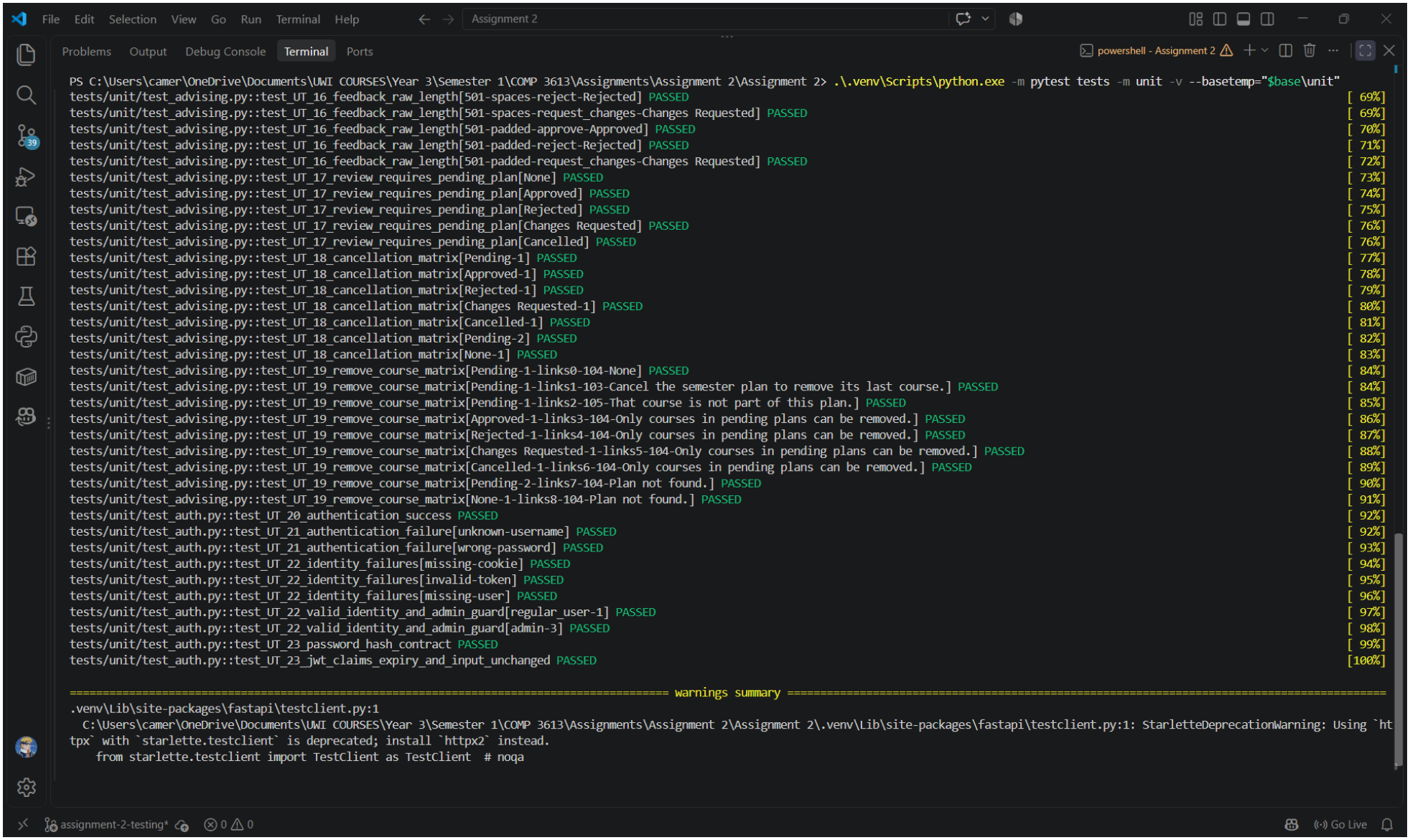


Figure 8.2. Genuine terminal capture Screenshot 2026-10-26 231742.png; 26 October 2026. Unmodified actual command/environment or passed variants; the final result is in Figure 8.3. Historical source attribution remains qualified.

8.1 Unit terminal evidence — completed result

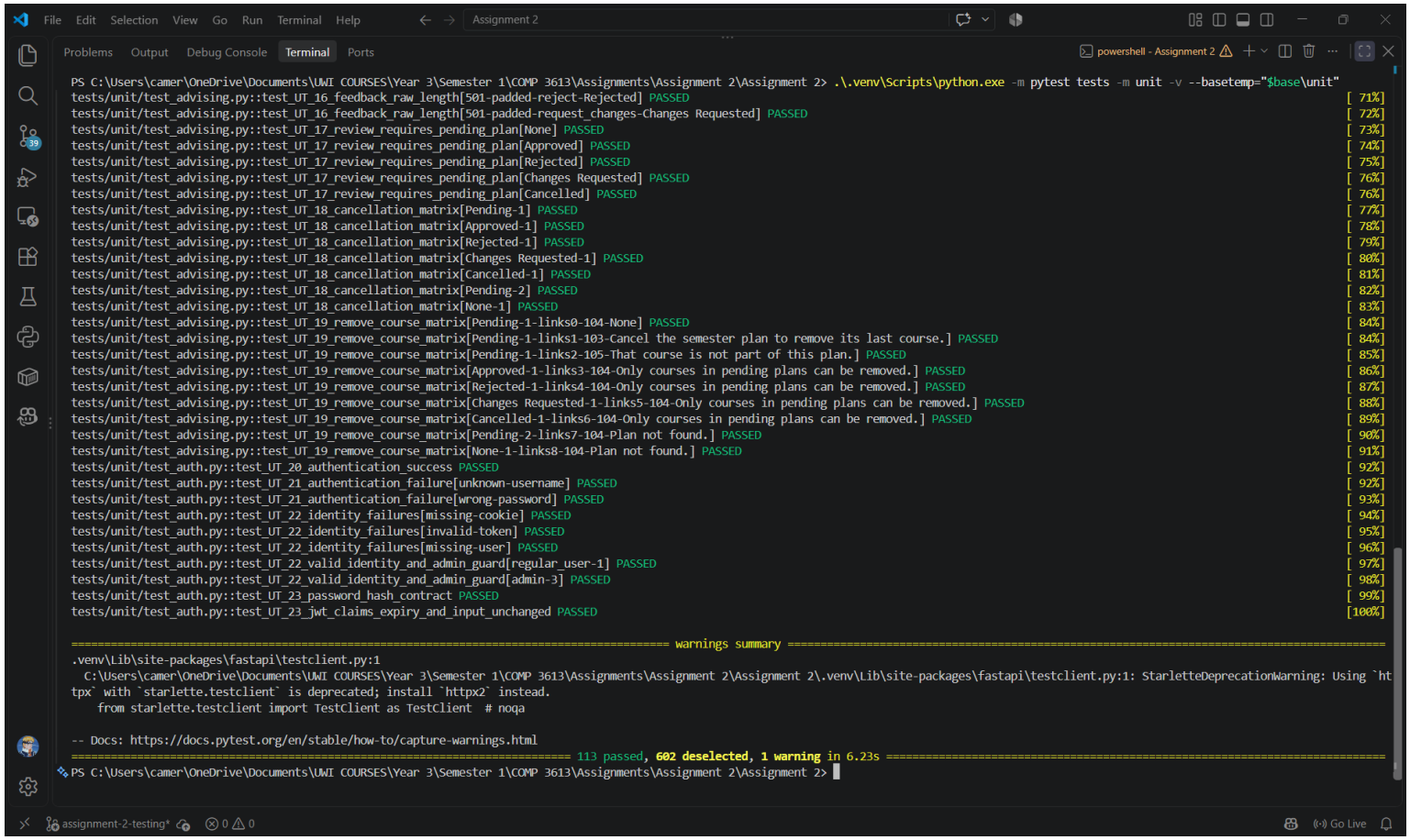




Figure 8.3. Genuine terminal capture Screenshot 2026-10-26 231747.png; 26 October 2026. Full image plus enlarged original result strip: 113 passed, 602 deselected, 1 warning in 6.23s. Historical source attribution remains qualified.

8.2 Integration Test Report

Objective: verify that real authentication, role/ownership dependencies, HTTP parsing, Student/Advisor workflows and database persistence work together. Components include TestClient, FastAPI, services/repositories and per-case SQLModel/SQLite with independent request sessions; database-only cases reach real constraints and rollback.

Environment: Windows/win32, Python 3.14.0, pytest 9.1.1, pluggy 1.6.0, anyio 4.15.1, pytest-asyncio 1.4.0 and pyproject.toml in the repository .venv. Separate actor clients perform real login and cookie handling. Exact latest marked dependency inventory/source hash and numeric exit-code record are missing.

Actual command (same writable temporary base defined in the unit start capture):

```
.\.venv\Scripts\python.exe -m pytest tests -m integration -v --basetemp="$base\integration"
```

Observed summary: 715 collected, 194 selected, 521 deselected; **194 passed, 1 warning in 30.52s**. Figure 8.6 shows the genuine terminal summary. The warning is the same disclosed TestClient/httpx deprecation. Latest complete marked text/XML is missing; a retained 59-pass Student-only run is narrower and is not substituted for 194.

Coverage/conclusion: queue receipt, approve/reject/request-changes decisions, feedback delivery, same-ID resubmission, removal, allowed/forbidden cancellation, replacement history, cross-user privacy, missing/invalid authentication, validation and fresh-session state assertions. All selected cases passed, but SQLite does not establish PostgreSQL, concurrency, deployed CSRF protection or human acceptance. The 20 correction variants were added later; their full-suite results are reported separately.

8.2 Integration terminal evidence — run capture 1

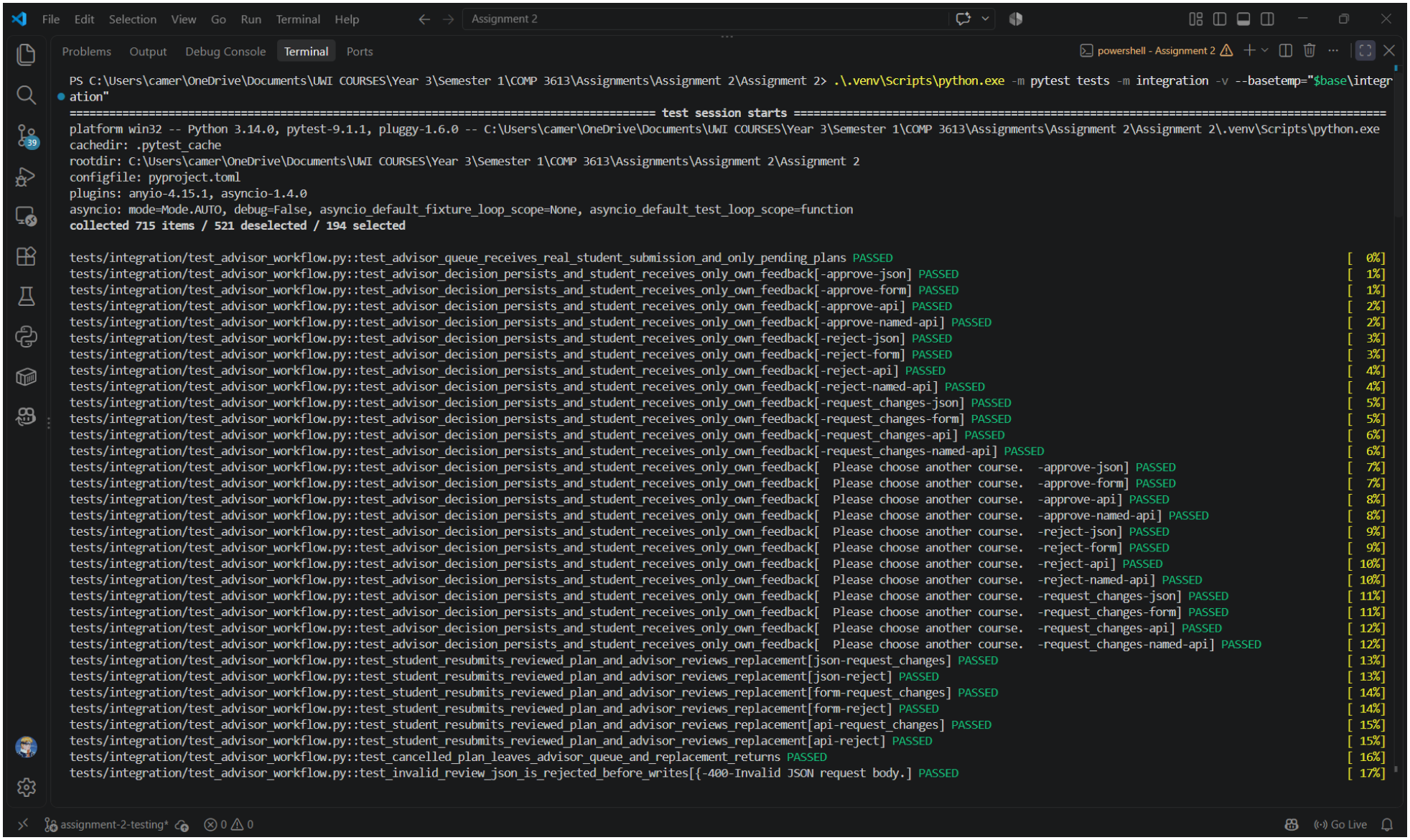


Figure 8.4. Genuine terminal capture Screenshot 2026-10-26 231956.png; 26 October 2026. Unmodified real Advisor/Student integration variants; final summary in Figure 8.6. Historical source attribution remains qualified.

8.2 Integration terminal evidence — run capture 2

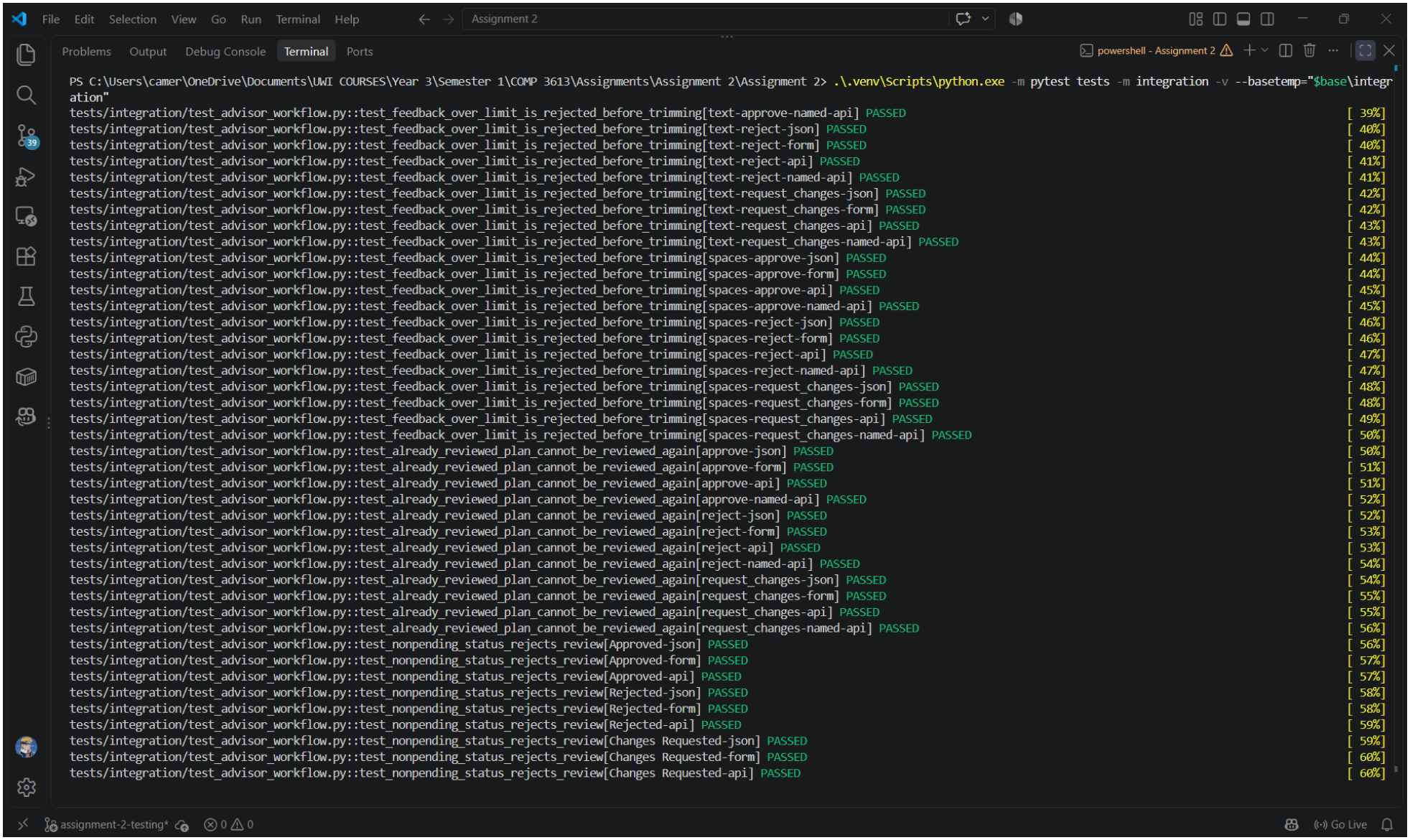


Figure 8.5. Genuine terminal capture Screenshot 2026-10-26 232025.png; 26 October 2026. Unmodified real Advisor/Student integration variants; final summary in Figure 8.6. Historical source attribution remains qualified.

8.2 Integration terminal evidence — completed result

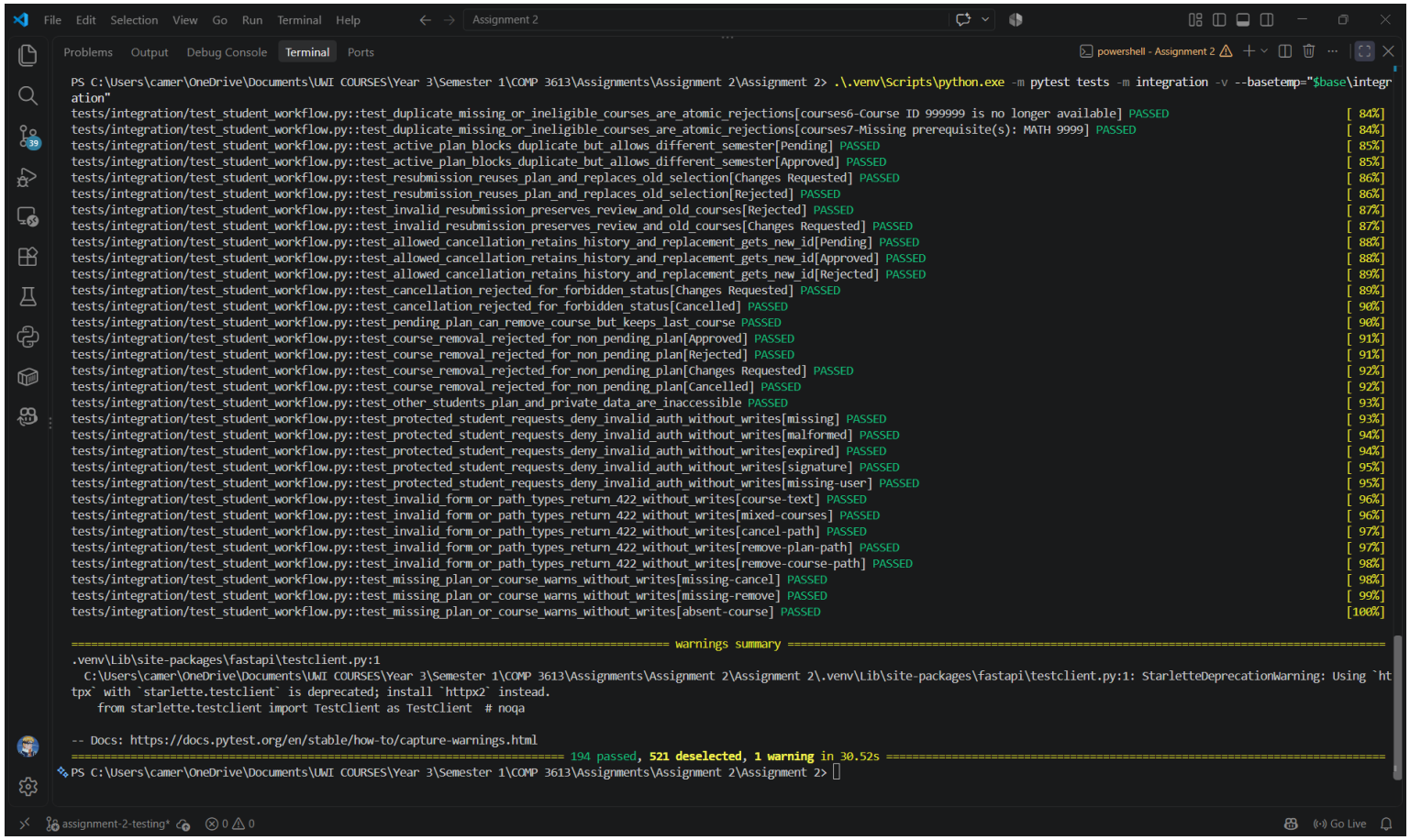




Figure 8.6. Genuine terminal capture Screenshot 2026-10-26 232058.png; 26 October 2026. Full image plus enlarged original result strip: 194 passed, 521 deselected, 1 warning in 30.52s. Historical source attribution remains qualified.

8.2.1 Supporting full-suite and corrected-build regression

Historical support: evidence/audit/pytest-output.txt, pytest-results.xml and pytest-run.json retain **715 passed, 0 failures/errors/skips, 1 warning, 76.25s, exit 0**. This broader run supports regression reporting and includes both marked groups; it is not a third category to add to 113 and 194.

Latest corrected source: evidence/corrections-2026-10-27/full-20261027T193800Z-7227960f.* retains **735 passed, 0 failures/errors/skips, 1 warning, 58.11s, exit 0**, started 27 October 2026 15:38 UTC−04:00. The metadata's 70 application/test SHA-256 entries were rechecked during assembly and all match the current files. Targeted correction record targeted-20261027T193352Z-b6385978.* separately retains 284 passes, 86.42s, exit 0; it overlaps the full suite.

Actual full-run command from retained metadata (absolute paths abbreviated with bracketed location placeholders; original command remains in the attachment):

```
python -m pytest tests -q -p no:cacheprovider
  --basetemp=[unique disposable path] --junitxml=[retained full-run XML]
```

Verbatim retained final text line, not a screenshot:

```
735 passed, 1 warning in 58.11s
```

Corrected environment metadata records FastAPI 0.141.1, SQLModel 0.0.42, SQLAlchemy 2.0.54, PyJWT 2.14.0, pwdlib 0.3.1 and SQLite 3.50.4. Runs used fresh writable temporary directories and no normal/private database; the unused bootstrap database was not created.

F01 HTML Student-role guards, F07 actual SQLite connection foreign keys and F08 registration boundary validation are fixed and covered locally. F06 computed contrast checks pass: normal table 14.81:1; hovered row 13.73:1; Approved paragraph 6.56:1. Human readability acceptance and corrected-build Postman/UAT checks remain pending. Old screenshots have not been relabelled as corrected-build proof.

Source: docs/assignment2/reports/audit-correction-report.md; evidence/corrections-2026-10-27/verified-summary.json.

Subsequent approved curriculum update (agent report, 27 Oct 2026): 36/18/12/18/9 = 93 non-overlapping credits; 46 affected tests passed; 751 full-suite tests passed with one warning, zero failures/errors/skips. The local database already matched and required no migration. No new Postman Runner or browser UAT run was performed. The 735-run text/XML above belongs to the earlier build.

8.3 Postman Collection Run Report

Collection: MyAdvisor - COMP 3613 Assignment 2, v2.1 export. **Objective:** exercise all 15 JSON operations and supporting real login/registration/logout through the dependent Student/Advisor plan lifecycle, including negative status/role/ownership requests.

Execution detail	Observed / qualified value
Evidence date	27 October 2026, user-confirmed; Runner displays 01:25:19 AM without an independent time-zone record.
Environment	Postman Collection Runner; UI shows environment none / No Environment; 1 iteration, duration 10s 697ms and average response time 29ms. Prescribed local isolated SQLite-backed setup is documented, but exact actual server URL, DB path, Postman version and tested build were not retained.
Requests / assertions	90 requests executed; **470 assertions passed**; **zero observed assertion failures**; **zero execution errors**. Six genuine captures collectively show requests 01–90; rows cut at scroll boundaries are not reconstructed.
Authentication/authorization	Separate Student/Advisor/independent Student login; expected anonymous and wrong-role denials; foreign ownership concealment and no-write read-back; response privacy.
Workflow scenarios	Progress/catalogue, invalid/valid submission, Pending queue, feedback limits, request changes/resubmit, reject/approve, immutable states, removal/cancellation/history/replacement and logout.
Conclusion	The recorded historical iteration passed all its assertions. A revised local JSON now has stronger requests 19/20 and passes static validation, but has not been run in Collection Runner. The category allocation discrepancy (F05) was resolved later; CSRF/signing-key and assertion-evidence limitations remain separately disclosed.

The following are genuine supplied Runner captures, not generated terminal text or reconstructed results. The exported JSON itself remains missing on GitHub despite being retained locally and attached to this PDF.

8.3 Postman Runner evidence — 1

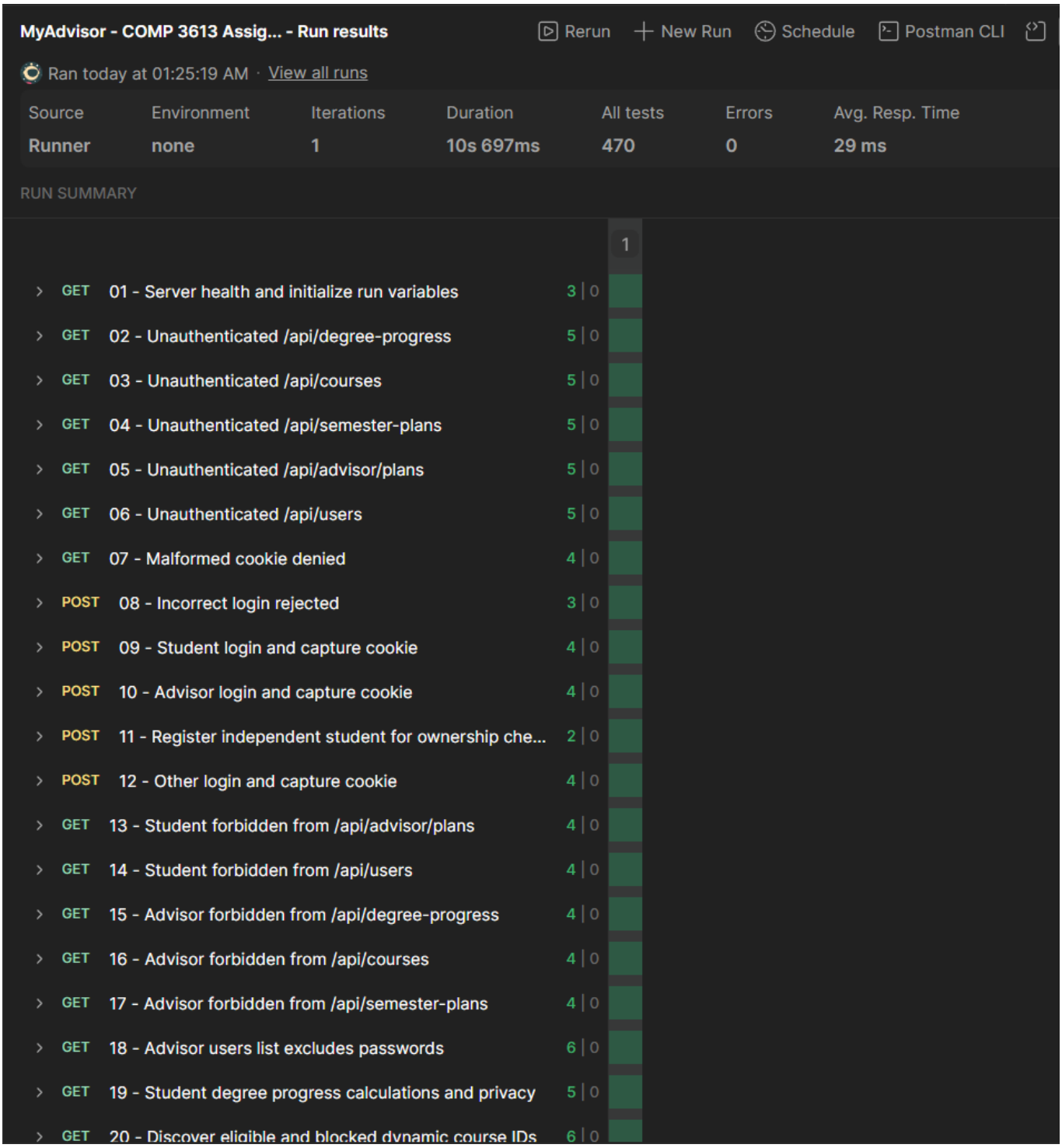


Figure 8.7. Genuine Runner capture Screenshot 2026-10-27 012608.png. Visible summary: 470 assertions, 0 errors, 1 iteration, 10s 697ms. The sequence collectively covers 90 requests; this is historical evidence. Unchanged crop of the run-results area; original file attached.

8.3 Postman Runner evidence — 2

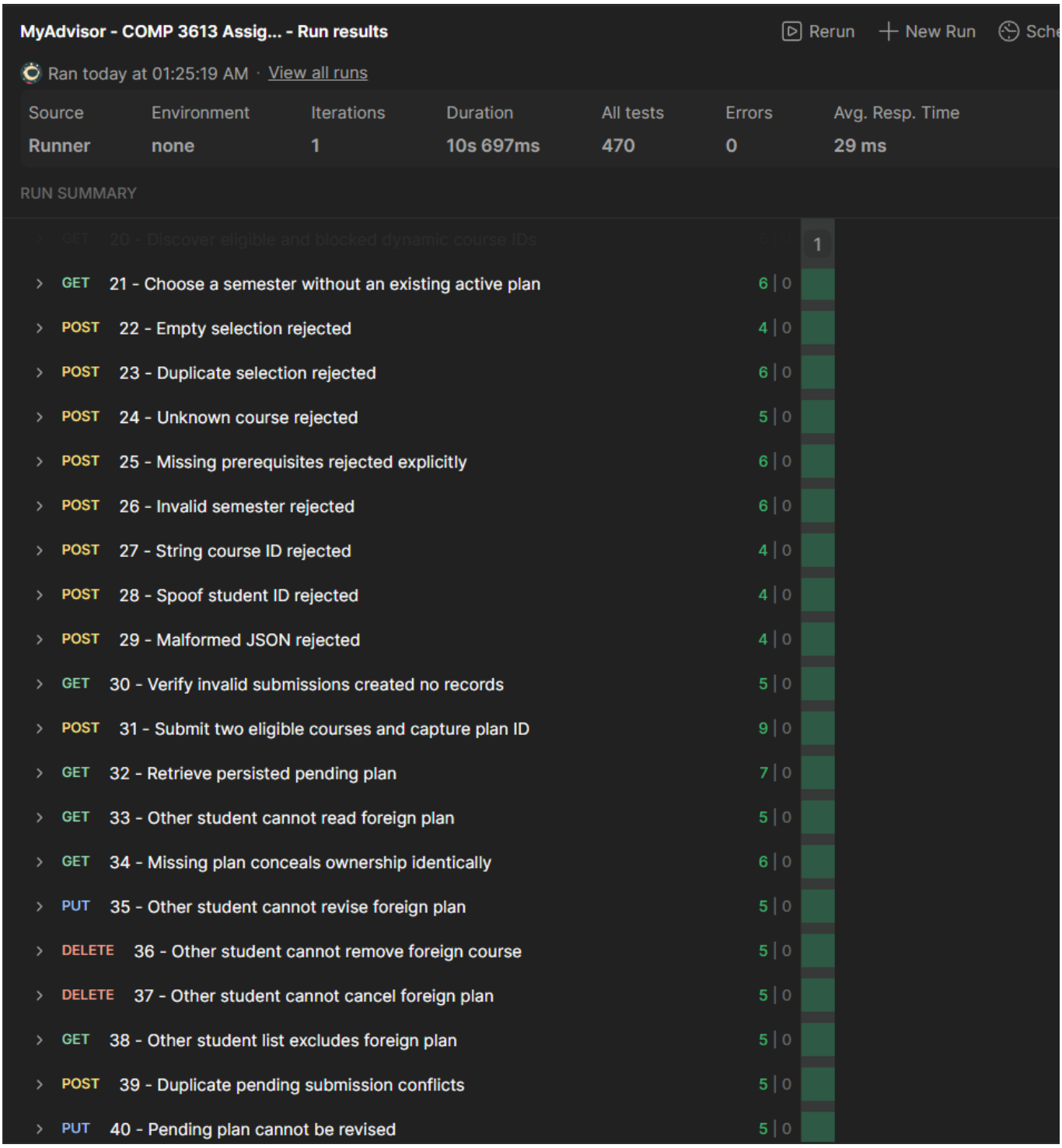


Figure 8.8. Genuine Runner capture Screenshot 2026-10-27 012643.png. Visible summary: 470 assertions, 0 errors, 1 iteration, 10s 697ms. The sequence collectively covers 90 requests; this is historical evidence. Unchanged crop of the run-results area; original file attached.

8.3 Postman Runner evidence — 3

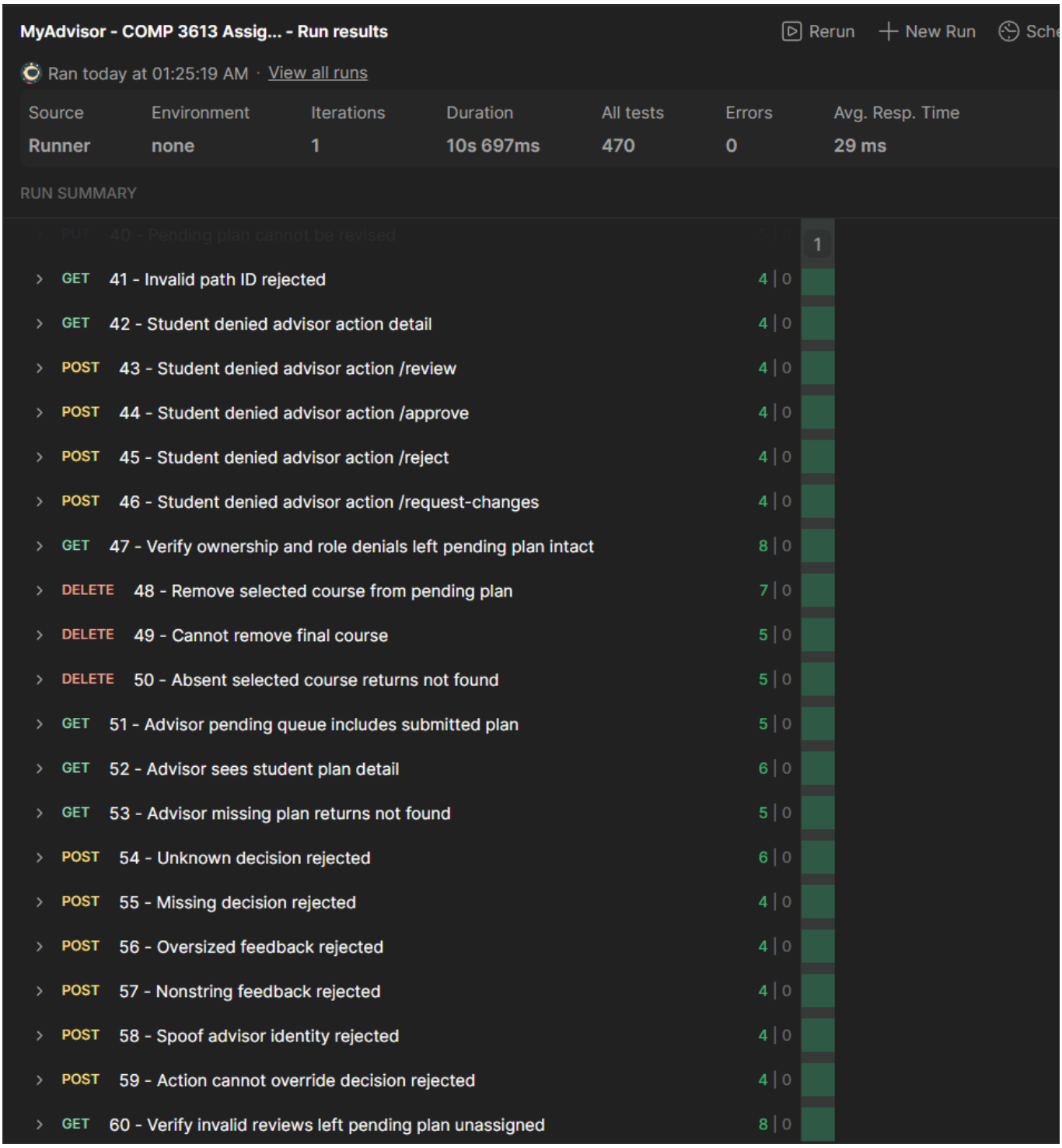


Figure 8.9. Genuine Runner capture Screenshot 2026-10-27 012658.png. Visible summary: 470 assertions, 0 errors, 1 iteration, 10s 697ms. The sequence collectively covers 90 requests; this is historical evidence. Unchanged crop of the run-results area; original file attached.

8.3 Postman Runner evidence — 4



Figure 8.10. Genuine Runner capture Screenshot 2026-10-27 012739.png. Visible summary: 470 assertions, 0 errors, 1 iteration, 10s 697ms. The sequence collectively covers 90 requests; this is historical evidence. Unchanged crop of the run-results area; original file attached.

8.3 Postman Runner evidence — 5

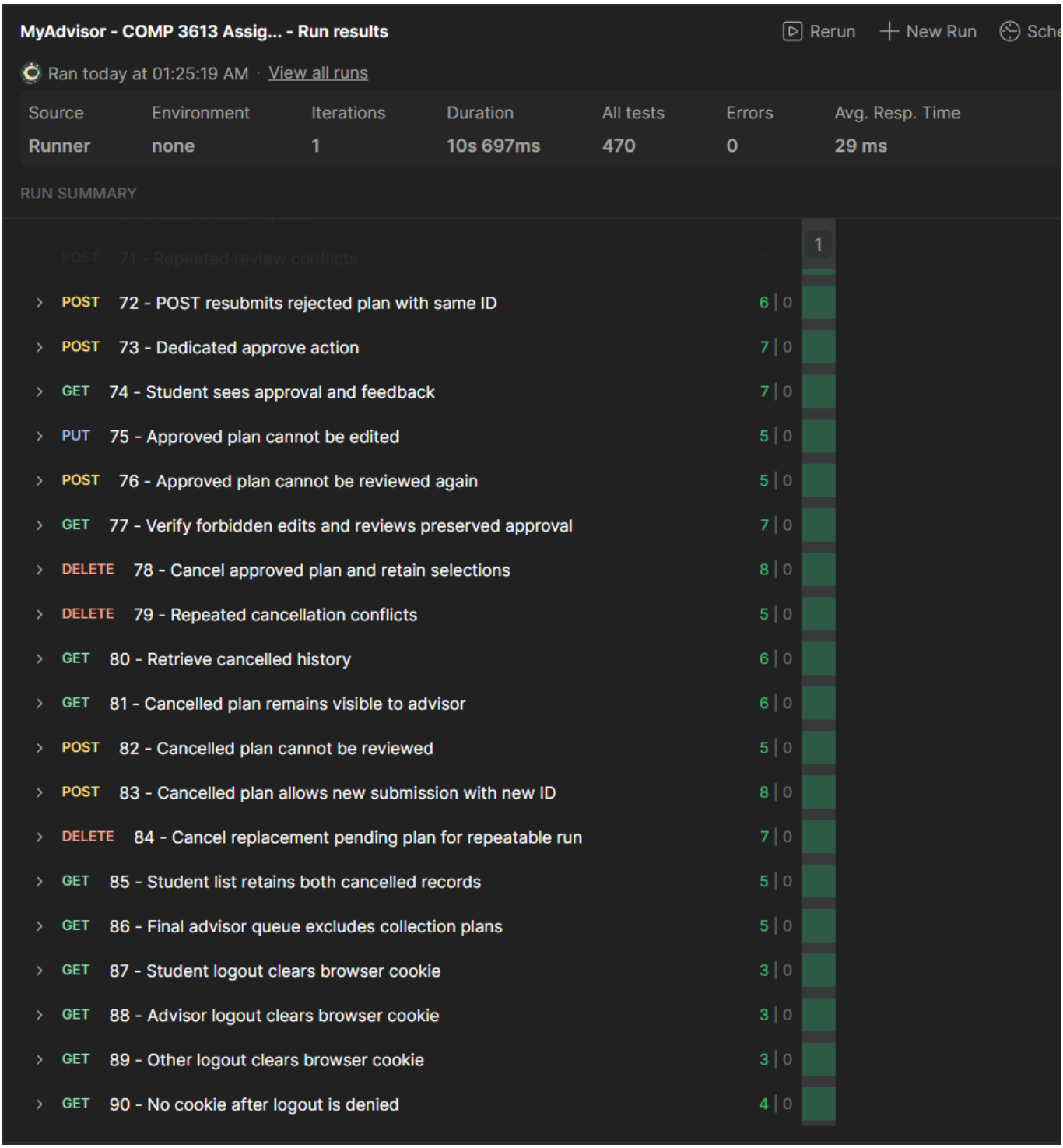


Figure 8.11. Genuine Runner capture Screenshot 2026-10-27 012839.png. Visible summary: 470 assertions, 0 errors, 1 iteration, 10s 697ms. The sequence collectively covers 90 requests; this is historical evidence. Unchanged crop of the run-results area; original file attached.

8.3 Postman Runner evidence — 6

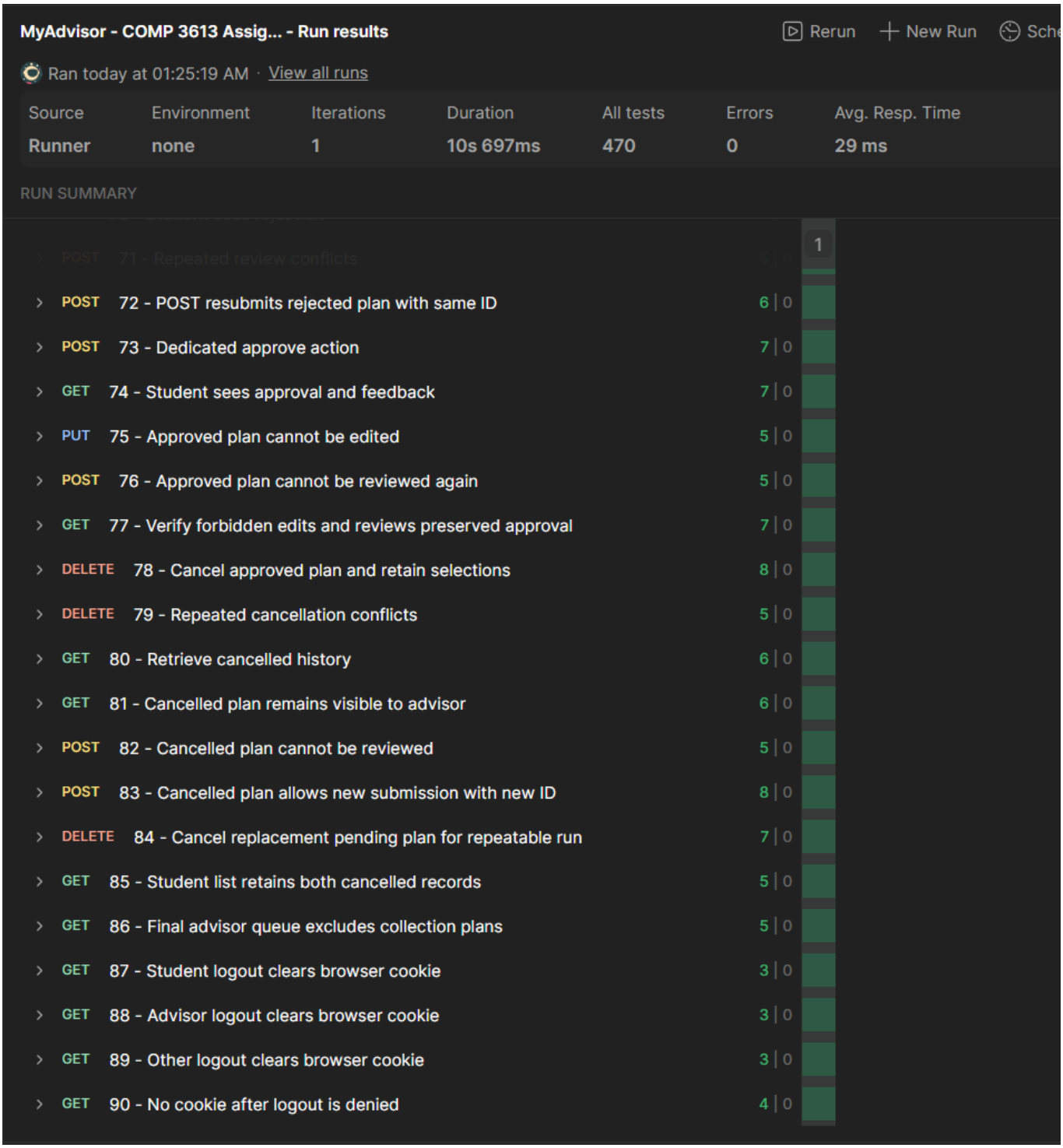


Figure 8.12. Genuine Runner capture postman-results.png. Visible summary: 470 assertions, 0 errors, 1 iteration, 10s 697ms. The sequence collectively covers 90 requests; this is historical evidence. Unchanged crop of the run-results area; original file attached.

Appendix A — UAT Execution Summary

Latest confirmed status: 1 Pass / 0 Fail / 19 Not Run. Tester Cameron Ramesar confirmed the 27 October 2026 execution/captures. Eighteen cases are partial; UAT-05 remains unverified. Pass applies only to UAT-06's recorded review objective with substituted data. No manual correction-stage execution is claimed.

Data deviation: actual initial A used MATH 1300 + MATH 2200 + MATH 2250 (3 courses / 9 credits); revised A used MATH 1300 + MATH 2200 + ENGL 1000 (3 / 9). The planned A was COMP 1201 + MATH 2250 (2 / 6), revised to COMP 1201 + ENGL 1000. Initial no-prerequisite choices do not demonstrate the planned nonempty-prerequisite path. Full actual-result fields and all 19 original UAT images are embedded as PDF attachments.

Case	Latest status	Observed evidence / remaining conditions
UAT-01	Not Run	Historical Degree Progress screenshot: 6 completed, 3 in progress, 84 remaining, 93 total and 6/93 (6%). Its older five category denominators added to 120. This was fixed in the approved fictional 93-credit configuration (36/18/12/18/9), but no new browser screenshot is provided. Still unverified: refresh persistence and current-build human acceptance. Evidence: UAT-02.png.
UAT-02	Not Run	Year 1 semester 1, math search and All departments are visible; MATH 1300, MATH 2200 and MATH 2250 are Eligible, each 3 credits; MATH 1200 is already completed. Remaining: Confirm semester switching, code and title searches, department and combined filters, no-match search and Clear Filters outcomes. Evidence: UAT-03.png
UAT-03	Not Run	Three eligible Mathematics courses are selected; selection count is 3. Completed-course restriction is visible. Post-submission image also shows in-progress and missing-prerequisite reasons, but that form is locked. Remaining: Confirm empty-selection prevention, pre-submission ineligible controls, actual count changes, review total and Back preserving choices; exact planned 0?2?1?2 sequence was not established. Evidence: UAT-03.png; UAT-04.png
UAT-04	Not Run	Submission success message; semester 1 Pending with MATH 1300, MATH 2200 and MATH 2250 (9 credits). Standard selection is locked; Remove and Cancel controls are visible. Remaining: Confirm refresh persistence and repeat submission prevention through the Pending form. Evidence: UAT-04.png
UAT-05	Not Run	No screenshot or supplied observation establishes the originally required Pending Home/Messages and sign-out/sign-in workflow. Later decision Messages are corroborating navigation evidence only. Remaining: Confirm Pending Home/Messages summary (actual 3 courses/9 credits) and selection retained after sign-out/sign-in. Evidence: None for the required workflow
UAT-06	Pass	Advisor Review shows the correct Student identity/ID/program, October 27 submission date, semester 1 Pending, MATH 1300/2200/2250, 3 courses and 9 credits, comments counter and all three decision controls. Reading the card leaves it Pending. Actual data satisfies the review objective with the documented substitution. Remaining: No remaining outcome check for this review objective with substituted data. Separate-profile setup/login attempts remain unverified under UAT-18/19; table contrast issue UI-01 remains open. Evidence: UAT-05.png

Case	Latest status	Observed evidence / remaining conditions
UAT-07	Not Run	Advisor Messages and Student Plan Courses show Changes Requested for semester 1 with identical feedback: "Replace MATH 2250 with ENGL 1000, then resubmit." Original three Mathematics selections remain selected in the editable form; update/resubmit instruction is visible and Cancel is absent. Remaining: Confirm Advisor action confirmation, removal from Pending queue, and Student Messages delivery (Student Plan Courses delivery is established). Evidence: UAT-07-request-changes.png; UAT-07-student-feedback.png
UAT-08	Not Run	Advisor Pending queue shows revised semester 1: ENGL 1000, MATH 1300 and MATH 2200; 3 courses, 9 credits. MATH 2250 was replaced. Empty Advisor comments input is visible; it does not independently prove stored feedback was cleared. Remaining: Confirm Student Pending state and review total for the actual 3-course/9-credit revision, current Pending Messages with prior feedback cleared, and exactly one revised Advisor queue entry after refresh. Evidence: UAT-08-resubmitted.png
UAT-09	Not Run	Student semester 1 is Approved; feedback: "Revised selection approved." ENGL 1000, MATH 1300 and MATH 2200 remain selected (9 credits); standard selection is locked, Remove absent and Cancel available. Student Messages in UAT-11 also shows this approval/feedback. Remaining: Confirm Advisor action confirmation and Advisor Approved feedback/queue removal; Student refresh persistence. Evidence: UAT-09-approved.png; UAT-11-rejected.png
UAT-10	Not Run	Student semester 2 submission success and Pending state with only MATH 2250, Industrial Statistics, 3 credits. Remaining: Confirm switching between semesters preserves each separate selection/status; later UAT-11 supports retained approval, not every switch. Evidence: UAT-10-second-plan.png
UAT-11	Not Run	Student Messages shows semester 2 Rejected with feedback: "Plan rejected for this UAT scenario; choose a different course mix." Semester 1 remains Approved with its approval feedback. Remaining: Confirm Advisor rejection confirmation/feedback and queue removal; Student rejected plan editability and Cancel availability. Evidence: UAT-11-rejected.png
UAT-12	Not Run	Advisor Review shows semester 3 Pending with COMP 1201 and MATH 2250, 2 courses and 6 credits. Remaining: Confirm Student Pending view and that A Approved/B Rejected were unchanged during this submission. Evidence: UAT-12-pending-plan.png
UAT-13	Not Run	"Course removed from the submitted plan." appears. Semester 3 stays Pending with only COMP 1201, 3 credits. No Remove button for the final course; "Cancel the plan to remove its last course." is visible. Remaining: Confirm dismissing Remove changed nothing and refreshed Advisor queue showed COMP 1201/3 credits. Final-course UI guard is established; no forged removal request is claimed. Evidence: UAT-13-course-removed.png
UAT-14	Not Run	Cancellation success; semester 1 Cancelled; "Revised selection approved." retained as feedback. Zero courses selected; new-selection instruction and eligible unchecked choice are visible; Cancel absent. Remaining: Confirm dismissing cancellation preserved approval, refresh persisted Cancelled, and Messages showed Cancelled. Evidence: UAT-14-approved-cancelled.png
UAT-15	Not Run	Cancellation success; semester 2 Cancelled; rejection feedback retained exactly. Zero courses selected, new selection possible and Cancel absent. Remaining: Confirm refresh/Messages Cancelled and A Cancelled/C Pending unchanged. Evidence: UAT-16-pending-cancelled.png

Case	Latest status	Observed evidence / remaining conditions
UAT-16	Not Run	Cancellation success; semester 3 Cancelled; zero selected courses; new-selection instruction and eligible unchecked choice visible; repeat Cancel absent. Remaining: Confirm Student Messages Cancelled, removal from refreshed Advisor Pending queue and preservation of other students if present. Evidence: UAT-15-rejected-cancelled.png
UAT-17	Not Run	After cancellation, semester 3 shows successful submission and a new Pending selection containing only ENGL 1000, Academic Writing, 3 credits. Old COMP 1201/MATH 2250 selection is absent. Remaining: Confirm Student refresh persistence and exactly one active new Advisor queue entry after refresh with ENGL 1000/3 credits. Historical Cancelled Messages entries are permitted, not a required additional check. Evidence: UAT-17-new-plan.png
UAT-18	Not Run	Student Home shows the seeded identity and Student navigation, no submitted semester plan and 0 updates. This is authenticated-state evidence, not a capture of the login attempts. Remaining: Confirm incorrect-login denial, valid-login destination, initial empty Messages/all A/B/C semesters and refresh retaining session. Evidence: UAT-01.png
UAT-19	Not Run	Advisor Review shows Dr. Williams and Advisor navigation, with no Student planning/progress links. This is authenticated-state evidence, not a capture of login attempts. Remaining: Confirm incorrect-login denial, valid Advisor Home destination and refresh preserving independent Student/Advisor sessions. Evidence: UAT-05.png
UAT-20	Not Run	401 Unauthorized / Session Expired page with Return to Login, plus an empty login screen. User attributes these to signed-out access restriction. No Student /admin 403 or Advisor restricted-tab fallback is demonstrated. Remaining: Confirm logged-in Student direct /admin 403, Advisor Student-tab fallback, unchanged queue, and fresh protected-page denial after each role signed out (which actor/URL for each). Evidence: UAT-20-access-denied.png; UAT-20-logged-out.png

Image mapping correction retained: UAT-16-pending-cancelled.png actually shows rejected semester B Cancelled and supports UAT-15; UAT-15-rejected-cancelled.png shows semester C Cancelled and supports UAT-16. Filenames/image bytes are preserved. No refresh, queue removal, unaffected-semester or chronology claim is inferred.

Browser/version, exact URL/DB and historical build remain unrecorded. New-build checks must cover missing login/logout, refresh/persistence, Student /admin 403, Advisor tab fallback, queue uniqueness/removal, feedback reset, corrected role guards/registration and human contrast acceptance.

Appendix B — Findings and Readiness

Evidence-qualified submission. The owner-approved fictional credit allocation now totals 93 and Codex reports a 751-pass full pytest suite. The historic 90-request / 470-assertion Postman run and 19 UAT captures are retained without implying retesting. The user will publish local changes manually. Remaining security/configuration and acceptance limitations are disclosed below.

Finding / priority	Disposition as of latest agent report
F01 High HTML role guard	Fixed locally; role/no-write tests passed in prior corrected suite. No new browser proof.
F02 High Postman export	Earlier JSON now exists on GitHub. Revised PM19/PM20 JSON not yet published/Runner-tested.
F03 High Git publication	Current corrected application, latest screenshots and reports still require manual Git update.
F04 High Manual UAT	Historical classification: 1 Pass, 0 Fail, 19 Not Run (18 partly observed). No new UAT.
F05 Resolved Credit rules	Approved fictional, non-overlapping 36/18/12/18/9 = 93. No destructive migration needed.
F06 Medium Contrast	Local style change and automated contrast checks reported; new human visual sign-off absent.
F07 Medium SQLite FKs	New app connections enable FK checks; earlier integration regressions passed.
F08 Medium Registration	Malformed email now returns controlled 422; earlier automated regressions passed.
F09 Medium CSRF	Open: authenticated HTML writes lack explicit CSRF/Origin policy. No live exploit claimed.
F10 Medium Provenance	Earlier 715/735 records and historical images retained; 751-run log not supplied to this PDF.
F11 Medium Assembly	Identity and lecturer-ordered 57-page document checked; this is an evidence-qualified copy.
F12 Low Documentation	Updated academic allocation reflected here; local/remote document versions may differ.
F13 Low Postman oracle	PM19/PM20 strengthened in local JSON; static check passed; new Runner still unexecuted.
F14 Low Repository hygiene	Duplicate Assignment 1 tree may confuse paths; no destructive cleanup attempted.
F15 Medium Signing key	Open: production signing-key strength/fallback not independently verified.
F16 Low Deprecation	One Starlette/httpx warning remains disclosed; not a failing assertion.

Submission note: Publishing the revised JSON and corrected code is a manual user action. Historical Postman/UAT results are genuine but do not verify the final corrected build. Do not convert partial UAT outcomes to Pass or claim live-production security clearance.

Appendix C — Compact Requirements Traceability

R01–R15 are existing local traceability IDs, not invented lecturer requirements. “Mapped” denotes scope; execution is governed by the qualified reports in Section 8 and Appendix A. The full retained matrix is attached.

Requirement	Unit / integration plan	UAT / Postman requests
R01 Degree totals/categories	UT-01–03; IT-02	UAT-01; PM19. F05 resolved; F13 rerun pending.
R02 Completed-only prerequisites	UT-04–06,11; IT-03,14	UAT-02–03; PM20,25.
R03 Valid nine semester choices	UT-07–08; IT-14	UAT-02,04,10,12; PM21,26,31.
R04 Nonempty/unique eligible selection	UT-09–11; IT-03,14,19	UAT-03–04; PM22–25,27–31.
R05 Plan lifecycle and duplicates	UT-12–13,17; IT-05–07,15,17	UAT-04–12; PM39–40,61–80.
R06 Advisor decisions / Pending review	UT-14–15,17; IT-05–07,13,17	UAT-06–09,11,20; PM51–78.
R07 Optional/500-character feedback	UT-14,16; IT-05–07,16–17	UAT-07–09,11; PM54–60,61,63,69,70,74.
R08 Returned-plan same-ID resubmission	UT-12–13; IT-06–07	UAT-08; PM65–68,72.
R09 Removal / last-course protection	UT-19; IT-10–11,18	UAT-13; PM36,48–50.
R10 Cancellation/history/replacement	UT-18; IT-08–09	UAT-14–17; PM37,64,78–86.
R11 Ownership/privacy/role isolation	UT-22; IT-02,11–13	UAT-20; PM13–18,33–38,42–47,81.
R12 Authentication/logout	UT-20–23; IT-01,12–13	UAT-18–20; PM02–12,87–90.
R13 Persistence / atomic rollback	UT delegation; IT-03–08,19–20	Lifecycle UAT/PM read-backs; fresh DB assertions.
R14 Validation/no partial writes	UT-08–11,15–17; IT-14–20	PM22–30,39–47,54–60,71,75–79.
R15 Public JSON API contracts	Additional API/security pytest files	All 15 JSON operations in PM01–90; Section 7.4.1.

Correction supplement: F01 HTML role guards, F07 SQLite engine integrity, F08 invalid-email protection and F06 contrast tests are documented in the earlier 735-pass suite. A subsequent 93-credit allocation update and stronger PM19/PM20 checks led to the agent-reported 751-pass run (46 affected tests). The revised collection remains unexecuted.

Appendix D — Lecturer Deliverables Checklist

This checklist preserves the lecturer's deliverable wording and order. Page references are filled from the final PDF's verified bookmarks in the accompanying checklist. "Included" describes document content; outstanding implementation/publication/acceptance requirements remain explicit.

Lecturer deliverable (exact wording)	Location / disposition
Name & ID	**Page 1.** Section 1 — verified name/ID.
Project Description	**Page 3.** Section 2 — included.
Use Case Diagram	**Page 4.** Section 3, Figure 3.1 — original included.
Entity Relationship Diagram	**Page 5.** Section 4, Figures 4.1–4.2 — original and readable keys included.
Application Wireframe	**Page 7.** Section 5, Figures 5.1–5.4 — original and three enlarged panels included.
Test Plans	**Page 11.** Section 6 — plans only; external template details unverified.
User Acceptance Tests Plan	**Page 11.** 6.1 — UAT-01–20 with actors, preconditions, data, steps, expected results.
Unit Tests	**Page 23.** 6.2 — UT-01–23 specifications and isolation/boundary strategy.
Integration Tests	**Page 27.** 6.3 — IT-01–20 specifications and corrected-build supplement.
Implementation	**Page 32.** Section 7.
Link to repository	**Page 32.** 7.1 — clickable intended repository; corrected publication incomplete.
Unit Tests	**Page 32.** 7.2 — actual pytest files, fixtures/mocks and code example.
Integration Tests	**Page 33.** 7.3 — TestClient, isolated DB, actors/auth and workflow example.
API Tests (Postman collection exported to JSON in repository)	Page 34. Local JSON attached; earlier 90-request JSON confirmed on GitHub. Latest strengthened JSON requires manual publication.
Reports	**Page 36.** Section 8.
Unit Tests Report (screenshot of test results)	**Page 36.** 8.1, Figures 8.1–8.3 — genuine 113-pass marked run.
Integration Tests Report (screenshot of test results)	**Page 40.** 8.2, Figures 8.4–8.6 — genuine 194-pass marked run.
Postman Collection Run report screenshot	**Page 45.** 8.3, Figures 8.7–8.12 — genuine 90-request, 470-pass Runner evidence.

Outstanding verification / manual actions: publish latest corrected code, revised Postman JSON and evidence to the intended GitHub repository; document the historical nature of original Runner/UAT results; decide whether to accept partial UAT evidence; independently verify production CSRF and signing-key settings. The owner-approved 93-credit allocation is resolved. No new browser or Runner execution is claimed.

Supporting files: use the PDF Attachments panel for original screenshots, exported JSON, complete UAT records, plans, traceability, full audits and genuine test logs/XML/source hashes. Attachments do not replace repository publication.